



National University of Computer and Emerging Sciences, Lahore



MoveSure

Tayyab Nadeem 21L-5830 BS(SE)

Muhammad Abdullah 21L-6101 BS(SE)

Aaon Raza 21L-6055 BS(SE)

Supervisor: Aleena Ahmad

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Abstract

This project aims at developing an AI-based solution referred to as "MoveSure" that helps people in maintaining correct posture and movement within workplaces, educational setups, and other institutions wherein extended periods of sitting are unavoidable.

This system captures human posture and movement and provides real-time alerts through camera technology to ensure healthy conduct.

Moreover, MoveSure gives users detailed reports that help them detect and correct poor postures. This serves to prevent the development of health problems such as back pains and musculoskeletal disorders harmful to the productivity and well-being of the people. Organizations can depend on MoveSure in ensuring their members make proper postures, thereby enhancing their physical health and performance over time.

Executive Summary

The MoveSure project is aimed at meeting the need for improved ergonomics in working places, schools, and universities since more individuals spend their majority of time sitting in front of computers or engaged with any other sedentary activities leading to increased musculoskeletal disorders (MSDs) due to bad habits in posture. MoveSure is fighting this issue through a new AI-based system that uses cameras as a means of monitoring people's postures and adjusting them in real-time. Good sitting habits will be cultivated from the users, reducing chronic long-term health issues that would be associated with back pain or posture complications.

The core function of MoveSure centers on the detection of incorrect posture and sending real-time alerts to the user, who would be immediately suggested to take corrective measures. Users receive immediate feedback on their posture, while the system produces detailed reports that track performance and trends in posture over time. Such data can be used by individuals and organizations alike in creating healthier work and learning environments. In addition, this system can generate reports for companies on comprehensive organizational posture habits with areas of improvement to extend the overall productivity.

Technically speaking, MoveSure has been engineered to be efficient and scaleable. This system uses AI camera technology to monitor users without requiring them to wear any special equipment or use hard-wired hardware, so it can be easily deployed in an incredibly wide variety of environments. And features like posture reminders which have adjustable timers and individualized reports mean that the platform is uniquely adaptable to the varied preferences of users and the different needs of organizations. The algorithm is designed to give instant feedback without much latency so that it can still continuously perform at optimal levels even during peak usage. Additionally, the system has a strong focus on security to ensure confidentiality of the data that may be made available by the users. MoveSure's performance ensures the users receive timely alert notifications and correct posture evaluations without manifesting noticeable delays even when dealing with tens of thousands of users at a time. The outcome for MoveSure is a reduced risk of MSD, along with better posture and healthier lifestyle in general. By providing people with actual, real-time monitoring of their posture- backed up by actionable feedback, MoveSure allows for proactive solutions to widespread concerns, highlighting how people and their organizations may better habitually maintain their posture-and, indeed, long-term health outcomes, too. The scalability of this project in various environments, coupled with its flexibility in accommodating different user demands, makes it a robust solution to achieve well-being in today's increasingly sedentary society.

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Chapter 1 Introduction

An average worker in offices, institutions, and other places spend a lot of time sitting down, leading to bad posturing and walking habits. This has been studied for various health conditions including musculoskeletal disorders as well as chronic back pain since it will not only discomfort a person but even pave way to a decrease in productivity. The idea of our AI-based project "MoveSure" will be to develop a complete platform that promotes healthier postures and movement behaviors while it continuously and in real time monitors them and provides feedback regarding posture and movement patterns. The camera technology involved with our system will analyze and assess posture and movement patterns and alert the user about if corrective action is required . Using granular reports, MoveSure will provide users with instructions on how to develop healthier habits over time. The system will allow the organization to foster a lifestyle culture of health because it provides recommendations on specific means employees or students can improve their physical fitness. In addition, MoveSure will act as a powerful tool for creating healthy workplaces through ergonomic means and productivity improvement.

1.1 Purpose of this Document

The main purpose of this project is to help people maintain better posture and movement habits, especially in places like offices or schools where they sit for long periods. MoveSure is designed to be an AI-powered assistant that keeps an eye on how you're sitting or moving and gives gentle reminders to help you avoid common issues like back pain. By offering timely alerts and easy-to-understand reports, it aims to encourage healthier habits, ultimately boosting overall comfort, productivity, and well-being in everyday environments.

1.2 Intended Audience

Intended audience of the project are organizations, businesses, educational institutions, and individuals who desire better health and productivity by ensuring proper posture and movement habits. Especially valuable for office workers, students, and teachers with long hours of sitting in a desk, such users are at an increased risk of developing posture-related problems in the near future. For the health and safety professionals, HR departments, and wellness coaches, the MoveSure system can be a tool that helps these people in the workplace to introduce healthier practices. The system would also be of benefit to any person who wants to improve their physical well-being as they receive real-time guidance and feedback regarding posture.

1.3 Definitions, Acronyms, and Abbreviations

List of all important definitions, acronyms, and abbreviations used in this document.

1.3.1 Definitions

- **Artificial Intelligence (AI):** A branch of computer science that focuses on building systems capable of performing tasks that typically require human intelligence, such as decision-making, pattern recognition, and learning.
- **Posture Detection:** The process of using sensors or camera technology to monitor and assess a person's sitting or standing posture.
- **Musculoskeletal Disorders (MSD):** A group of conditions that affect muscles, tendons, ligaments, and nerves, often resulting from poor posture or repetitive strain.
- **Inertial Measurement Unit (IMU):** A device that measures and reports the velocity, orientation, and gravitational forces of a user, often used in systems to monitor movement.
- **Real-Time Feedback:** Immediate responses generated by the system to inform users of incorrect posture. This feedback is crucial for corrective mechanisms.
- **Performance Tracker:** A feature that tracks the user's productivity in relation to their posture, providing insights and statistics based on posture habits.
- **Admin:** Refers to administrative users of the system who have special access privileges, such as the ability to generate reports, monitor multiple users, and modify system settings.
- **Report Generation:** A system feature that creates summaries of user posture data and productivity, providing insights into user habits and organizational trends.

1.3.2 Acronyms and Abbreviations

- **CNN:** Convolutional Neural Network
- **LLM:** Logical Layered Model
- **API:** Application Programming Interface
- **AI:** Artificial Intelligence
- **IMU:** Inertial Measurement Unit
- **MSD:** Musculoskeletal Disorders

- **SDG:** Sustainable Development Goal
- **FYP:** Final Year Project
- **MVP:** Minimum Viable Product
- **UI:** User Interface
- **UX:** User Experience
- **Agile:** Agile Development
- **SCRUM:** Scrum Development
- **REST:** Representational State Transfer

1.4 Conclusion

The section introduced the MoveSure project by stating its core objectives; which focused on health problems in posture through AI-powered monitoring and giving feedback real-time. The system is going to boost healthier habits while at the same time improving general productivity in workplaces as well as educational settings. It was developed to help encourage healthier habits while improving overall productivity in workplaces and educational settings. The system is made to promote healthier habits that enhance productivity in the workplaces as well as educational settings. This system is aimed at improving healthier habits while increasing productivity through general enhancement of activity in both workplaces and educational institutions.

Moving forward, this paper considers the vision and requirements of the project, a comprehensive review of relevant literature and technologies, and detailed breakdown of the system's design and architecture-areas which represent core features of MoveSure: technical implementation, scalability, and security.

Chapter 2 Project Vision

This project aims to establish a healthier and even more productive environment by providing an AI-driven system to be able to foster proper posture and movement behaviors in the workplace, schools, and other places where the experience is mostly based on extended sitting. MoveSure aims to be a reliable solution to help individuals avoid posture-related health issues through real-time monitoring, alerts, and comprehensive feedbacks. In this regard, by embracing this tool, organizations are in a position to create a culture of well-being while reducing the risk of those problems identified with musculoskeletal disorders and generally improving on productivity and the quality of life of all people involved.

2.1 Problem Domain Overview

MoveSure addresses common issues caused by bad postures and sitting for long hours in the workplace and learning institutions, causing health problems to be around issues with back pains and loss in the efficiency of work output. This technology uses AI cameras scanning users' real-time movement and posture with detecting incorrect alignment or inactivity. MoveSure sends immediate alerts to user once it identifies an issue, prompting them to take corrective measures. Moreover, it includes graphical reports in detail that can be found on the MoveSure website, showing posture patterns and suggesting which areas should be improved to facilitate better habits and reduce health risks due to poor posture in individuals and organizations.

2.2 Problem Statement

Poor postures resulting from the prolonged sitting in workplaces and educational establishments lead to many health consequences and lower productivity. It is a practice with very few solutions that monitor and correct the habits in real time while providing detailed reports for the user to track progress. Such an absence poses challenges in maintaining healthy ergonomic practices for both the individual and organizations, which may augment the risk of long-term health risks and reduce efficiency.

2.3 Problem Elaboration

The problem with individuals is that most spend long hours in workplaces and educational environments with poor sitting postures, which result in health problems and decreased productivity. The MoveSure aims to curb this by offering a system that monitors and alerts its users of any wrong posture, hence promoting healthier habits.

The secondary challenge is how to assist the user in correcting his posture. MoveSure meets this with the production of full and real-time reports on the website, thus affording the user an opportunity to track progress and patterns of unhealthy behavior.

Other problems that would be dealt with include user engagement and privacy. To avoid misuse or neglect of the system, the proposed solution includes alerts and reminders at real-time to prompt corrective actions. For the issue related to the privacy concerns, all video data captured is processed confidentially in a way that only posture-related information will be used for the solution while keeping the identity of the person as confidential. The solution to the main problem and issues associated with it assures overall effective and safe solutions.

2.4 Goals and Objectives

The goal of this project is to develop an AI-driven solution that makes life easier for people by maintaining a good posture and avoiding health risks that are usually carried forward from hours of sitting. For this purpose, it also plans to deliver an online platform that the users can utilize to get statistics and reports so that they can understand how to improve their posture over time. Furthermore, it will use an engaging alert system that encourages better ergonomic practices in workplaces and educational institutions. Finally, MoveSure will ensure privacy through a secure interface while giving actionable feedback for a general sense of well-being that entails continued posture improvement.

2.5 Project Scope

As previously discussed, the project will include a website where people, organizations, and institutions can all be brought together to improve ergonomic practice. The system will constantly monitor the real-time posture and movements of the users through AI-based camera technology. The website will also include reports in which the posture patterns are generated with detailed graphical reports so that users can find an understanding over time and improve their habits. It will also have an alerting system that alerts the users to take corrective action when they remain inactive for long durations. Lastly, MoveSure will provide analytics in terms of privacy-protected data, which would enable organizations and their users to apply healthier ergonomic practices while safeguarding personal data security.

2.6 Sustainable Development Goal (SDG)

The following are the sustainable development goals that this project aims to achieve.

2.6.1 SDG-3 (Good Health And Well-Being)



Figure 2.1: SDG Goal 3

This project falls under one of the major sustainable development goals, "Good Health and Well-being." Many working places and learning centers have a low level of productivity due to bad postures and other sedentary activities which later will lead to health problems. Real-time postural guidance will be offered through the application of MoveSure to these individuals, and they have the ability to lessen such health-related issues. As a result, they will achieve good well-being-the base of good productivity and efficiency in their daily activities.

2.6.2 SDG-9 (Industry, Innovation, and Infrastructure)



Figure 2.2: SDG Goal 9

Another secondary SDG is "Industry, Innovation, and Infrastructure." In the use of AI technology, MoveSure offers a more innovative way of overcoming ergonomic challenges. This is health, in terms of habit and practice, to modern, health-based representations that better arm organizations with new practices that can lead to a much more sustainable and productive working environment.

2.7 Constraints

The key constraints of this project are that the system cannot initially check the correctness of the user's posture or movements because the analysis is AI-based, whereas users need time to get adjusted and possibly will not be getting perfect immediate feedback, consistent usage will be required to get accurate improvement. Moreover, the system will perform effectively in environments that have good lighting and are also camera-angle friendly. Some users would therefore have to go through various levels of malfunction due to inadequate lighting or positioning. In addition, the generated reports for posture improvement are limited only to the metrics and insights available from this system, and the users may depend only on these insights without guidance on improvements.

2.8 Business Opportunity

This will provide a great business opportunity to all the stakeholders who would be a part of this project. Offices and educational institutions can benefit from higher productivity and reduced health-related issues among employees or students, thereby having lesser sick days and more efficiency. Moreover, MoveSure can be added to the wellness programs of the software houses as a new tool for encouraging the health of the employees. They can further boost their image as a healthy-conscious employer through this value-added service. Organizations implementing MoveSure will also experience improved overall performance and a healthier workplace environment, thus yielding long-term savings on healthcare costs. The project itself offers a chance for profits since the site might charge subscribers or license the technology to other business companies that will enhance the health and productivity levels of workers, thereby creating a source of recurrent revenue.

2.9 Stakeholders Description/ User Characteristics

The primary stakeholders of MoveSure include:

- Office Workers and Remote Employees
- Educational Institutions (Students, Teachers, and Administrators)
- Employers and HR Managers
- Health and Wellness Consultants

2.9.1 Stakeholders Summary

The key stakeholders of the MoveSure system are office workers, remote employees, and organizations that aim to encourage a healthy work environment. Office workers and remote employees are the primary users, who use the system to monitor and improve their posture in long hours spent at workstations. They will reduce the risk of acquiring MSDs since they will be given real-time feedback and guidance in improving their posture.

Educational institutions, and the students, teachers, and administrators who make up the fabric of them, also benefit from MoveSure, since it encourages healthier habits in the classroom or the remote learning environment. Organizational stakeholders also include the employers and HR managers; they are the ones responsible for making sure that the employees have access to the system and data to better fortify productivity at work by keeping health-related absenteeism low. Through MoveSure, health and wellness consultants can offer comprehensive analytics and insight into workplace ergonomics, further supporting the organization in providing wellness programs tailored to its workers' needs.

2.9.2 Key High-Level Goals and Problems of Stakeholders

The core objective for the stakeholders of the MoveSure project will be to realize several high-level objectives by using AI-powered posture monitoring to enhance workplace health and productivity. A system that promotes better ergonomic practices and minimizes the risk of posture-related health problems can eventually contribute to overall better workplace well-being. Though an improvement in a healthier work environment will ultimately reduce the number of MSDs that develop with increased productivity, it is the improvement in well-being that benefits organizations.

Key concerns for stakeholders are easy integration of AI posture monitoring system into the existing workplace setting without disturbing normal working processes. The second concern is user compliance with correction measures activated by the system. Stakeholders are also concerned with maintaining user privacy, ensuring the system adapts to various body types and environments, and providing real-time feedback that is accurate and easy to act upon.

2.10 Conclusion

MoveSure is the AI-based solution for posture and productivity changes in workspaces and education sectors with real-time monitoring, feedback, and detailed reports for better ergonomic practices support key sustainable development goals-good health and innovation. Despite some constraints, the project presents significant business opportunities by enhancing employee well-being, and increasing efficiency.

Chapter 3 Literature Review / Related Work

This chapter examines existing research and technologies in posture monitoring and ergonomic assessment. It highlights advancements in AI, wearable sensors, and camera-based systems, identifying gaps and insights to guide the development of MoveSure for promoting healthier ergonomic practices.

3.1 Definitions, Acronyms, and Abbreviations

List of all important definitions, acronyms, and abbreviations used in the literature reviews.

3.1.1 Definitions

- **Ergonomics:** The study of designing workplaces and products to fit the users, aimed at maximizing comfort, efficiency, and safety.
- **Musculoskeletal Disorders (MSD):** A broad term referring to injuries or disorders that affect the human musculoskeletal system, particularly those arising from repetitive or improper posture.
- **Forward Head Posture (FHP):** A postural issue where the head is positioned forward in relation to the body, leading to strain on the neck and shoulders.
- **Posture Asymmetry:** The uneven distribution of weight or improper alignment of the body, often resulting in strain and long-term physical issues.
- **Inductive Textile Sensor:** A sensor integrated into textiles used to monitor changes in posture by detecting physical pressure and movement.
- **Convolutional Neural Network (CNN):** A deep learning algorithm particularly effective in analyzing visual data, commonly used for image recognition and video processing in posture detection.
- **Electrocardiogram (ECG):** A test that measures the electrical activity of the heart over a period of time and can be used in monitoring physical health during posture assessment.
- **Inertial Measurement Unit (IMU):** A device that measures the specific force, angular rate, and sometimes the orientation of the body, often used in posture monitoring systems.
- **Asymmetry Detection:** The process of identifying uneven or incorrect body alignment, which can lead to physical strain or injury.
- **Long Short-Term Memory (LSTM):** A type of recurrent neural network (RNN) used to process sequences of data, often employed in real-time posture detection systems to track movement over

time.

3.1.2 Acronyms and Abbreviations

- **AI:** Artificial Intelligence
- **CNN:** Convolutional Neural Network
- **ECG:** Electrocardiogram
- **IoT:** Internet of Things
- **FHP:** Forward Head Posture
- **IMU:** Inertial Measurement Unit
- **LSTM:** Long Short-Term Memory
- **MSD:** Musculoskeletal Disorder
- **WMSD:** Work-Related Musculoskeletal Disorder
- **ML:** Machine Learning
- **3D:** Three-Dimensional

3.2 Detailed Literature Review

3.2.1 Intelligent Posture Training: Machine-Learning-Powered Human Sitting Posture Recognition Based on a Pressure-Sensing IoT Cushion (MDPI, 2022)

The study involves the use of a specific hardware to monitor posture.

3.2.1.1 Summary of the research item

This paper[1] presents a smart posture training scheme using machine learning algorithms with a pressure-sensing IoT cushion in order to recognize and improve human sitting posture. The proposed system copes with health problems caused by long periods of poor sitting posture through actual real-time monitoring and feedbacking. The system derives pressure data from the cushion by employing supervised machine learning models that classify 15 different sitting postures with accuracy over 98.82% and also identifies six seated stretches with an accuracy of 97.94%. The authors derive real-world validation for its performance over five different real-world workplace environments and lay out training strategies for the machine learning models. Additionally, it suggested the presence of data-driven stretch recommendation feature aligned with the practice of physiotherapy standards to establish good posture habits

among its users.

3.2.1.2 Critical analysis of the research item

The study effectively demonstrates the possibility of IoT with machine learning for monitoring and improvement in sitting posture. High accuracy rates in posture and stretch recognition represent robust performance of the model. A valuable strength of this research is its validation across multiple real-world settings, which enhances the applicability and generalizability of the system. Incorporating a recommendation system about stretch adds value in terms of indicating poor postures as well as giving possible solutions for action.

This dependency, though on the pressure-sensing cushion, involves a couple of drawbacks. First, the mere existence of a particular cushion will discourage widespread application due to cost, availability, as well as practicality across diverse environments. Also, differences in users' BMI or sitting patterns may affect the precision of the system because it would be sensitive to those differences and possibly interfere with the pressure reading. Divergence in users' BMI significantly influences the recognition of posture as observed by the study and would be useful to adjust personally. The system provides timely feedback but does not discuss long-term user engagement strategies that are necessary for ensuring sustained improvements in posture.

3.2.1.3 Relationship to the proposed research work

The study is directly applicable to the MoveSure project as It seeks to use an AI-powered system for encouraging proper posture and movement behavior. MoveSure can derive lessons from the high accuracy that was achieved through these machine learning models used for posture recognition. Unlike this pressure-sensing cushion, MoveSure uses camera-based technology, thus eliminating special hardware requirements, making it scalable, and easy to deploy both in offices and educational institutions.

In this regard, the challenges that the study establishes—such as the BMI impact on accuracy—indicate a call for models that, by their very nature, can include the individual differences in physiological make-up, a factor that MoveSure can integrate into its AI algorithms. On the other hand, the stretch recommendation does the job of integral feedback and actionable insight provided to the user by MoveSure. It is therefore possible to improve engagement among users and popularize healthier ergonomic practices if built going forward with such integration. Such studies highlight the prospects associated with real-time monitoring and individualized feedback inherent in the objectives of MoveSure.

3.2.2 ERG-AI: Enhancing Occupational Ergonomics with Uncertainty-Aware Machine Learning and Large Language Model Feedback (Springer, 2023)

This study too uses specific hardware, however goes more into the model that it uses.

3.2.2.1 Summary of the research item

This paper[2] introduces the topic of ERG-AI, a comprehensive machine learning pipeline to augment occupational ergonomics with predictions of worker postures from wearable sensor data and encompasses estimation of uncertainty for accuracy and reliability of predictions. ERG-AI makes use of the availability of large language models (LLM), such as GPT-4 for translating posture prediction along with uncertainty into understandable occupational health risk assessments and personalized recommendations. It is aimed at the prevention of work-related musculoskeletal disorders(WMSDs) through actionable worker and decision-maker insights built from sensor data. The pipeline ingests data, preprocesses, trains models, and deploys, all targeted to efficiency and reduction in carbon footprints. Real-world datasets have been applied to evaluate the system's performance in real-time posture prediction and ergonomic risk assessment.

3.2.2.2 Critical analysis of the research item

The ERG-AI is crucial in occupational ergonomics and develops integration between machine learning and LLMs towards highly interdisciplinary complete ergonomic assessments. Moreover, uncertainty-aware models increase the reliability of the predictions in terms of worker health decision-making. Using LLMs translates user-friendly feedback that fills the gap between complex data analysis and practical application.

However, reliance on wearable sensors also raises some concerns regarding user compliance and data integrity. The workers would be perceived as wearing a sensor, which would be intrusive, or they forget to put one on, which would not provide full collections of data. The effectiveness of the system relies significantly on the quality and continuity of the sensor data, which is likely to vary in real-world settings. Its intensity in computing resource consumption in processing large datasets and complex models like LLMs might limit the accessibility of such systems to smaller organizations with limited technical infrastructure.

3.2.2.3 Relationship to the proposed research work

ERG-AI is highly relevant to the goals of developing an ergonomic practice with AI-based solutions by MoveSure. The way ERG-AI incorporates uncertainty estimation may benefit posture detection

algorithms of MoveSure to be more accurate and reliable. LLMs for the generation of individualized ergonomic recommendations share with MoveSure the same goal- one of full feedback provision to users.

Over the user compliance issues identified during ERG-AI, the MoveSure system avoids such issues by applying a camera-based system rather than wearable sensors. High-probability continuous data collection is therefore allowed by keeping cameras monitoring posture without requiring active participation from the users. User engagement and possibly prolonged behavioral changes can also be augmented by employing real-time risk assessments and personalized feedbacks in relation to recommendations through AI language models. In terms of emphasis on energy efficiency as well as scalability, ERG-AI offers needed knowledge to MoveSure when implemented in an optimized way for wide adoption.

3.2.3 A Novel Smart Chair System for Posture Classification and Invisible ECG Monitoring (MDPI, 2023)

A study which makes use of a chair system to achieve posture monitoring.

3.2.3.1 Summary of the research item

This paper[3] provides an intelligent chair system designed to monitor posture while sitting and conducting latent ECG monitoring. The intelligent system under the seat incorporates load cells to measure weight distribution and conducts on armrests made with conductive materials to acquire ECG signals. Machine learning models are trained in the classification of five to seven sitting positions with accuracy levels over 97.4% for five-class models and 87.5% for seven-class models. The ECG monitoring module uses single-lead ECG to monitor heart activity without any explicit awareness on the part of the user. Here, methods of outlier detection are applied to improve the quality of the ECG signal where segmentation performance was 88.21% and accuracy at 90.50%. The system maintains the working conditions optimum, which should be with proper posture and good cardiovascular condition without interference with any other activity.

3.2.3.2 Critical analysis of the research item

This paper combines posture monitoring with cardiovascular health assessment to provide a comprehensive approach to enlist the possibility of using load cells and models with ML in order to improve posture classification accuracy. A very innovative concept about invisible ECG monitoring without causing discomfort or intervention for continuous health assessment.

The scalability and accessibility of this system are not good because the hardware used has to be

specialized-the smart chair equipped with sensors. These smart chairs may be too expensive and unaffordable for many organizations to replace their existing chairs. Additionally, the system is limited to an examination of posture only while sitting in a chair and excludes posture or movement when the users are away from the chairs. The monitoring of ECG, although useful, will raise legitimate privacy issues about the collection of very sensitive health information without the express awareness of the users.

3.2.3.3 Relationship to the proposed research work

The way a smart chair system classifies posture aligns with the objective of MoveSure-to promote posture through real-time monitoring and feedback. MoveSure can take inspiration from the high accuracies achieved in posture classification by using different machine learning models. Nevertheless, AI-powered cameras by MoveSure give a flexible and cost-effective solution that lets rid of the need for specialized furniture, thus allowing monitoring posture in all sorts of settings, not just when users are seated on particular chairs.

This also allows MoveSure to consider elements of health monitoring such as photo capture for signs of fatigue or stress visual that are typically intended to provide a more holistic well-being solution. It is also critical to maintain a focus on the privacy considerations of the smart chair system and thus, the data security and consumer consent become highly important while designing MoveSure, especially when personal health information is being created and processed.

3.2.4 Human Posture Recognition Using Machine Vision for Work-from-Home Environments

A paper which utilizes a smartphone camera to observe posture.

3.2.4.1 Summary of the research item

This paper[4]q proposes a system of posture recognition in work-from-home environments by applying machine vision. The authors employ a small-scale CNN by utilizing a smartphone's built-in camera for the classification of proper and improper sitting postures. A custom dataset is utilized, which calculates distances and angles between body joints for classifying sitting posture and then trained the model. The system achieved high accuracy, in which it detected proper posture 85.18 percent of the time and improper posture with an accuracy of 92.07 percent. The study goes to show that real-time monitoring of posture can be realized without expensive equipment using affordable smartphones.

3.2.4.2 Critical analysis of the research item

It does present a promising and practically feasible solution for the improvement of good ergonomic practices in remote working settings. It is highly accessible and scalable with a smartphone camera-based CNN. However, the small-scale CNN dependency of the study may be limited in scalability, especially when postures get complicated or when the number of users increases. Furthermore, this work was limited to only work-from-home conditions that, of course do not fit well into all other conditions that exist. For instance, office working environments come with different types of ergonomics challenges which are not covered in the dataset. Also, while the dataset has been contributed by physical therapists, some varieties of conditions may be limited to not satisfy all possible varieties of human anatomy and posture types.

3.2.4.3 Relationship to the proposed research work

The proposed study of MoveSure shares similarities with this paper, mainly in its objective of monitoring and enhancing the posture of the user with the help of computer vision. Like the study, MoveSure will use machine vision and deep learning models for posture analysis. However, beyond sitting posture, the research in the name of MoveSure proposes dynamic, real-time posture correction through standing and walking activities. The other reason this study targets work-from-home environments, and MoveSure intends to look at a wide user base, from office and educational departments, will require adopting the model to function ideally in a diverse environment with additional feedback systems to engage and sustain the habits for behavior changes in the users.

3.2.5 Modelling Proper and Improper Sitting Posture of Computer Users Using Machine Vision (MDPI, 2023)

A study that practices portability using a smartphone and simple CNN.

3.2.5.1 Summary of the research item

This paper[5] introduces a computer vision-based system for monitoring and classifying proper and improper sitting postures is proposed in this paper, in the context of a work-from-home scenario. The model makes use of the smartphone camera to track body points and estimate angles between joints, which are then classified into proper or improper postures. In order to show the possible high accuracy of posture recognition up to 92.07%, a small-scale convolutional neural network was developed. Such a study shows how the system can be applied in rehabilitation and ergonomic assessment of long-term sitting postures.

3.2.5.2 Critical analysis of the research item

A simple CNN and smartphone camera gives the system a high degree of access and ease of deployment; this is a strength, which becomes even stronger when translated into work-from-home setups. The model's posture classification delivers a high degree of accuracy; however, one can only speculate about the overall system in possibly challenging environmental conditions or in the presence of various body types. In addition, it can appropriately detect instances of bad posture, but does not give all the features of proposing corrective suggestions or providing real-time feedbacks and, hence, would not prove useful as a tool for a complete improvement in posture.

3.2.5.3 Relationship to the proposed research work

The current study comes directly in alignment with the goal of MoveSure that has to do with the real-time monitoring and correction of posture. Using low-cost, widely accessible technology, such as smartphone cameras, MoveSure can use a similar approach, but with posture tracking, it adds value with this model by providing real-time corrective feedback as well as the generation of general reports that will be interactive to help in the long-term correction of posture. Adding the angle of ergonomic analysis, MoveSure would appeal more strongly to an organization application.

3.2.6 Spatio-Temporal Calibration of Multiple Kinect Cameras Using 3D Human Pose Estimation (MDPI, 2022)

A posture monitoring paper using multiple Kinect cameras.

3.2.6.1 Summary of the research item

This paper[6] proposes a technique for real-time calibration of multiple Kinect cameras to capture 3D human poses. Information from several cameras is fused together to minimize the probability of occlusions and enhance human body tracking accuracy. This approach does not require expensive equipment, such as light probes or checkered patterns, but only relies on motion of a human body within the scene for on-the-fly calibration. The Kinect skeleton-tracking system analyzes asynchronous 3D pose data from the cameras; the data is also aligned in order to achieve accurate skeleton tracking. This system is very important for applications that need continuous tracking of human motions, such as rehabilitation and ergonomic assessments.

3.2.6.2 Critical analysis of the research item

This makes the study highly relevant to fields such as health care and ergonomics with its innovative usage of multiple Kinect cameras and reducing occlusion and enhancing 3D pose estimation. This means calibration is easier without requiring special equipment, something that means an important improvement in relation to traditional approaches and makes the system more accessible and flexible.

3.2.6.3 Relationship to the proposed research work

This study directly addresses the objective of MoveSure in offering posture monitoring using non-invasive camera-based solutions. Just like the Kinect system, MoveSure may employ several cameras to refine posture detection without much occlusion. While MoveSure gives more emphasis on single-camera configurations for wider-range usage, the multi-camera calibration method proposed in this paper may be potentially used for environments requiring higher precision, such as busy working spaces or industrial environments. Last of all, the approach taken by the study to track human motion without the need for any sort of specialized calibration tools resonates with the goal of developing an easy-to-use and deployable system for ergonomic monitoring in different settings, towards which MoveSure strives.

3.2.7 Real-Time Human Skeletal Posture Estimation Using mmWave Radar (IEEE, 2020)

A study making use of radar technology to achieve posture analysis.

3.2.7.1 Summary of the research item

The study[7] introduces a novel system, mm-Pose, built on millimeter-wave (mmWave) radar technology that can identify skeletal posture of people in real time. It is different from those more common vision-based systems which have direct visual data for detecting the human skeletal joints and tracking how they move in their environment; instead, mm-Pose uses radar signals to detect human skeletal joints and track their movements, especially useful in low-light environments or situations where the user is partially occluded. It follows 15 skeletal joints, from the head to the neck, and shoulders, to limbs and provides full-blown visualization of the posture adopted by the user in real-time. The system has been relatively applied in healthcare, ergonomics, and safety aspects, where continuous posture monitoring is key.

3.2.7.2 Critical analysis of the research item

The mmWave radar-based approach also has other advantages over camera-based systems, as it potentially can be used in locations where visual information is lacking, such as dimly lit rooms or over

extremely crowded workspaces. Because the system does not use cameras to track skeletal joints, it may be less invasive and could be more secure, because all identifiable visual information is avoided. However, one significant limitation of mm-Pose is its requirement of complex and expensive hardware that constitutes a piece of radar, so that therefore it may limit the accessibility of it to the consumers. The system mostly focuses on the skeletal joints; therefore it cannot capture essential posture aspects such as muscle strain and soft tissue alignment.

3.2.7.3 Relationship to the proposed research work

As with MoveSure, the goal of the mm-Pose system is to monitor posture in real-time, though mm-Pose uses a camera-based system. Potentially one benefit that MoveSure may have over mm-Pose is that it is cheaper to implement and deploy since no specific hardware, such as a radar sensor, is needed. However, perhaps the main strength of mm-Pose-such as its functionality under poor lighting or occluded conditions, for example-can be an advantage to MoveSure in that its own system might be extended in robustness to changing conditions. Building from AI-driven camera-based solutions, MoveSure actually provides an arguably more scalable and accessible posture correction alternative across different settings.

3.2.8 IoT System for Real-Time Posture Asymmetry Detection (MDPI, 2021)

This particular study resorts to the use of pressure sensors to observe human posture.

3.2.8.1 Summary of the research item

This paper[8] proposes an innovative postural asymmetry detection system based on real-time IoT technology integrated with textile pressure sensors. Ten measurement points were applied on both the seat and backrest of a chair to recognize seven different sitting positions based on machine learning-based classification where fuzzy logic was applied, for instance, in mild, moderate, or severe cases. The system offers real-time visual feedback through a mobile application, helping them monitor the kind of posture set during work hours. This is to prevent long-term musculoskeletal problems due to the asymmetrical sitting positions maintained for a long period of time.

3.2.8.2 Critical analysis of the research item

This paper does a good job of explaining how IoT and pressure sensors can be applied to detecting and adjusting posture asymmetry, which is one of the most common causes of back pains in the office environment. This fuzzy logic algorithm seems well suited for dealing with inherent variability in seating positions, but its practical applicability is restricted to furniture systems with an embedded pressure sensor, making it somewhat less scalable and not very practical for widespread usage. Further, the

system leaves no immediate intervention like alarms or haptic feedback for further improvement of the level of user engagement and quicker corrective actions.

3.2.8.3 Relationship to the proposed research work

This study is applicable to MoveSure as it points out a real need for real-time feedback of users on their posture. Although MoveSure uses camera-based AI instead of pressure sensors, understandings from this paper on how asymmetry can be detected can improve upon tracking and correcting bad postures within a system. With visual and possible haptic mechanisms of giving feedback, MoveSure can better facilitate engagement by users to remediate poor postures over time.

3.2.9 Real-Time Forward Head Posture Detection and Correction System Utilizing an Inertial Measurement Unit Sensor (MDPI, 2023)

This study utilizes a wearable sensor to track head posture.

3.2.9.1 Summary of the research item

This paper[9] introduces a real-time system for detecting and correcting forward head posture (FHP), which is commonly caused by prolonged use of smartphones and computers. The system uses an inertial measurement unit (IMU) sensor placed on the user's cervical spine to track head position. This system has been able to process the sensor data in order to detect transitions into FHP and has given visual alerts encouraging the correction of posture to the users. In experiments conducted in a real-world office environment, FHP detection was completed within 0.5 seconds, and the system was successful in improving the craniovertebral angle for all participants over time.

3.2.9.2 Critical analysis of the research item

This study has provided a solution toward solving FHP, an everyday problem in working environments. The system takes advantage of the use of less intrusive hardware in the form of IMU sensors, making it lighter, more portable, and less expensive than similar hardware. Still, this system mainly concentrates on head posture while not touching other major points such as shoulders and the spine, which are all impacted by bad postures. Moreover, dependency on wearable sensors could degrade user compliance with such devices in the long run.

3.2.9.3 Relationship to the proposed research work

This research can be considered as one start and can be utilized in the MoveSure project by incorporating relevant techniques that may be able to detect some forms of poor postures, such as forward head posture.

Though camera-based systems are utilized by MoveSure, the alert mechanism in this paper will be highly effective to immediately warn users whenever they slouch or carry some other forms of postures. Hence, this research strengthens the concept put forth at the heart of the MoveSure design-that of timely feedback.

3.2.10 A Deep-Learning Based Posture Detection System for Preventing Telework-Related Musculoskeletal Disorders (MDPI, 2021)

A study that discusses the use of multiple CNNs achieve accurate and efficient real-time posture analysis

3.2.10.1 Summary of the research item

This paper[10] covers an MSD prevention system by means of deep learning-based posture detection toward prevention of MSDs in a work-from-home setting, thus in telework conditions, mainly escalated during the COVID-19 pandemic. The system depends on a CNN to process video in real-time and classify incorrect postures of the neck, shoulder, and arms. The risk of serious long-term health problems is also minimized because the system provides recommendations to the teleworkers to correct their posture with low power consumption and real-time performance at 25 frames per second. The system shows over an 80 percent accuracy with posture detection.

3.2.10.2 Critical analysis of the research item

The study introduces a new solution to the ergonomic problems of remote work, especially in the context of the pandemic. It makes use of CNNs that will enable accurate and real-time posture detection, avoiding any need for any specific hardware, thus making it accessible to a wide variety of users. However, the system's focus is limited to a small number of body parts (neck, shoulders, arms), potentially overlooking the impact of full-body posture on MSDs. Furthermore, while the system offers real-time feedback, its accuracy of 80

3.2.10.3 Relationship to the proposed research work

This research actually aligns with the goals of MoveSure because both target the promotion of healthier posture practices in workplaces. While move sure focuses on generalized postures correction using artificial intelligence, the methods based on deep learning involved in this paper provide insight into how neural networks may be incorporated into move sure for the purpose of enhancing the accuracy of real-time detection as well as feedback. In addition, the know-how about real-time feedback mechanisms can be applied to promote healthier behaviors among teleworkers and thus propel MoveSure into a higher thrust both in the office and at home.

3.3 Literature Review Summary Table

Table 3.1: Literature Review Summary

Application	Features	Relevance to Application	Limitations
Intelligent Posture Training[1]	Machine-learning-powered posture recognition with pressure-sensing IoT cushion	Posture recognition for improved ergonomics and health through AI-based feedback	Requires specialized hardware (pressure-sensing cushion)
ERG-AI[2]	Uncertainty-aware machine learning and large language model feedback for ergonomics	Provides personalized ergonomic recommendations using AI for real-time posture feedback	Wearable sensors may affect user compliance and data integrity
Smart Chair System[3]	Load cell-based posture classification with invisible ECG monitoring	Combines posture monitoring with cardiovascular health assessment	Expensive and limited to sitting posture, raises privacy concerns for ECG data
Wearable Sensors and AI[4]	CNN-based posture recognition using a smartphone camera to detect sitting posture	Directly applicable to MoveSure's posture monitoring feature for real-time feedback on mobile devices	Limited to sitting posture detection and may not scale for more dynamic postures
Modelling Proper and Improper Sitting Posture[5]	Machine vision-based system for posture classification using smartphone cameras	Uses machine vision for affordable posture correction and ergonomic assessments	Limited to sitting posture and lacks real-time corrective feedback
Spatio-Temporal Calibration with Kinect Cameras[6]	Multiple Kinect cameras for real-time 3D pose estimation	Real-time 3D human posture tracking for ergonomic assessments with multiple cameras	Relies on multiple cameras for accurate tracking, may require complex setups

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Application	Features	Relevance to Application	Limitations
Real-Time Skeletal Posture with mm-Pose[7]	mmWave radar-based skeletal posture estimation in real-time	Non-visual real-time skeletal posture tracking for low-light or occluded environments	Requires radar hardware, making it less accessible for general use
IoT Posture Asymmetry Detection[8]	IoT with pressure sensors and fuzzy logic for posture asymmetry detection	Real-time asymmetry detection to enhance posture monitoring accuracy	Limited scalability and practical use in broader environments
Real-Time Forward Head Posture Detection[9]	IMU-based forward head posture detection with real-time feedback	Focuses on detecting specific issues like forward head posture in real time	Limited to head posture, does not cover full-body alignment
Deep-Learning Based Posture Detection System for Telework[10]	Uses CNNs to detect neck, shoulder, and arm postures in real-time and offers corrective feedback at 25 FPS	Provides real-time feedback for teleworkers to prevent MSDs caused by poor posture. Aligns with real-time posture correction goals of MoveSure	Focuses only on neck, shoulder, and arms, missing full-body posture impact and has only 80% accuracy

3.4 Conclusion

Advances in AI-based posture detection systems, which focus on optimization of ergonomic practices, especially at workplaces and telework arrangements, are known to be one key takeaway from the literature. An effective approach of real-time posture monitoring and feedback to undertake various studies using machine learning, deep learning, and IoT technologies has also been proved although certain limitations, like a narrow focus on the body and hardware dependency, can be seen. With this knowledge, it is possible to develop MoveSure with AI in real-time and inspire healthy posture habits in a more holistic and accessible way.

Chapter 4 Software Requirement Specifications

This chapter includes the important features of the project. It will include functional, non-functional requirements, assumptions, use case diagrams and multiple other subtopics involved in the project.

4.1 List of Features

- Tracking users' posture by integrating AI with camera technology
- Detecting wrong posture and sending a notification in real-time to alert the user
- Generating a report for the user detailing their productivity, posture, and performance
- Generate data reports for companies highlighting areas of improvement and overall statistical insights
- Adaptable to multiple environments including schools, offices, universities, etc.
- Step by step guide on how to improve overall posture and well-being
- Generate end of day reports for companies highlighting key statistics

4.2 Functional Requirements

This part describes functional requirements for various groups of stakeholders, such as Admins, Users, and the MoveSure system.

4.2.1 Functional Requirements for Admin

- The respective personnel should log on to the system using their credentials since such users have been assigned such roles.
- The application shall allow the admin to create more users.
- The application should enable the admin to delete or deactivate users in the application.
- The application should help an admin in updating any profile or account information pertaining to any user as well as user roles.
- The application should allow the admin to change any user for the sufficient user role available.
- The application should enable admin to obtain reports and analytics related to users' activities posture within system.

- The application should allow the admin to create hierarchies of the users and clubs by allocating different user types to users.
- The System shall allow the administrator to determine the acceptable level of user activity and how the notifications should be triggered if the threshold is not met.
- The System will allow the administrator to give general directions regarding the peripheral devices and any other technologically related apparatus to accompany the system.
- The System shall give control over the elements of the system which integrate such as cameras or light sensors.
- The application should provide the admin with reports and feedback on the status of different devices in the network as well as their diagnostic logs.
- The system shall also enable the administrator to view the past posture details of every user of the system.

4.2.2 Functional Requirements for Users

- The users shall be able to access their accounts.
- Users shall be able to pull the live posture information from the user dashboard. It will also be designed to raise live posture alerts and receive feedback from users.
- Users shall be able to choose their personal settings in the alert system (e.g. what kind of sounds or images or texts should be used).
- Users shall have access to the past activity log and posture status data of the users as well.
- Users shall be able to change their individual posture thresholds as well as the tips on how long they can afford to sit without any activity.
- Users shall be given in addition to general work ergonomics, pages that would offer personal recommendations to users on how they can effectively better their posture.
- Users shall have the opportunity to store posture evaluation data and reports for self-usage or for medical reasons.

4.2.3 Functional Requirements for System

- The system shall utilize video feeds that will track the movement of users in real time and will assess the users' degree of crouch.
- The system for active alarm and the triggering feedback seeks and evaluates posture disorders and

deficiencies of movement by means of external impact within a definite period of time.

- The system shall also have a provision for multi-user access and feedback with summons rooms for concurrent observation of the procedure.
- Posture correction aids including content detection systems have been interfaced and implemented to suppress deviations from an acceptable user posture as happened in that working day.
- Different types of reporting on the analysis of users' posture shall be done by individual users or system admins/teachers on daily, weekly, or monthly bases.
- The posture logs, the activity, and the reports must be stored in a database and there must be a proper system for the safekeeping of the posture logs, the activity, and the reports like for instance ideally in MongoDB.
- There should always be user privacy preserved in the sense that there shall be data encryption because there shall be a need for any data to be transmitted 'safely' and stored 'safely as well. For instance, systems, and institutions, which might find merit in integrating external applications or equipment, such as health-related wearables, would be facilitated by the system.
- The system shall also consolidate the hardware installation e.g. camera installation so that whatever posture that is being checked is done so to the actual precision i.e. down in performance.
- Looking at the deployment of the system as was earlier mentioned, the system shall be sufficiently liberating to permit either connection to the internet or local deployment hosted depending on the organization's needs.

4.2.4 Functional Requirements for MoveSure System Components

- All these activities concerning storing various records of a person's posture's history, any activity logs of the person and reporting of them shall be incorporated into the system which will integrate MongoDB with it.
- In this system recognized methods of AI algorithms will be employed specifically tensor flow and pytorch for their efficiency in the execution of a timely required process of detection of an optimal posture position.
- The system processes should be designed to use asynchronous processes in such a way that many requests can be satisfied without changing the day to day working of the system.
- All the hardware components and systems to which the camera and sensors are linked will be managed and controlled by the system together with the cameras and other sensors

4.3 Quality Attributes

- Performance: The system will give quick responses whenever incorrect posture is observed.
- Usability: The system will have a user-friendly interface ensuring smooth navigation.
- Reliability: The system will have minimal downtime to guarantee an authentic statistical report.
- Maintainability: The system will have cohesive architecture and clean code.
- Interoperability: The system will allow smooth integration of cameras and other hardware.

4.4 Non-Functional Requirements

List of the non-functional requirements of the project.

4.4.1 Reusability

The system will be coded in a way that keeps it modular, so that specific components such as posture detection, report generation, etc., can easily be reused across the different modules and parts of the project. Meaning when future features are to be integrated into the system, the team won't have to redo the previous codes.

4.4.2 Performance

The optimum handling of a large number of users who can be active at the same time will be implemented in the system. Real-time feedback for posture correction, without too much latency, ensures the user can make immediate adjustments to the corrected postures. The hardware resources such as CPU, Memory and Camera input processing will be kept well utilized in an efficient manner so that the high workload conditions do not go against the smooth performance.

4.4.3 Extensibility

The system will be designed in such a manner that allows easy extension of the new features to the system, such as fuller posture tracking or other kinds of reports, without actually changing any aspect of the system architecture. Thus, it will be able to develop with new technologies or requirements.

4.4.4 Security

User privacy will take center stage, and all data relevant to posture or user information will be encrypted at the time of proper storage within the system. Access control mechanisms will be applied so that such

sensitive information will only be accessed by authorized personnel-like administrators or managers of the company.

4.4.5 Accuracy

The system will accurately detect bad postures with the use of AI models that could be trained to handle cases wherein such types of subjects may exist together with the camera's angles and lighting conditions. It will try to minimize false positives or negatives to ensure that only genuine issues are reported and not alert users when there isn't one.

4.4.6 Usability

To make navigation intuitive for users, the system will also have a friendly user interface that clears a notion of posture correction in the form of feedback, showing easy access to reports, and has customizable settings. In several areas, the system will be operable with minimal training or guidance.

4.4.7 Reliability

The system will ensure consistent performance with minimal downtime, guaranteeing that posture monitoring and notifications function accurately and without interruptions. Mechanisms like backup systems and error handling will be in place to ensure that the system remains operational during failures. Logs will enable the immediate detection of problems occurring, hence enable prompt resolution, and there will be maintenance scheduled in a way to cause minimal user disruption.

4.5 Assumptions

- Users will have an active camera for posture monitoring.
- The system will have internet connection for feeding back in real-time, giving users access to their reports and syncing data.
- Users are expected to know how to interact with web applications and navigate the system's user interface.
- Organizations will provide sufficient lighting and appropriate setup for posture monitoring as the camera-based posture monitoring system assumes that the environment has adequate lighting and space for accurate posture tracking.
- Users will consent to share their posture and movement data for monitoring, reports, and alerts.

4.6 Use Cases

List of use cases.

4.6.1 User Login

Name	User Login
Actor(s)	User
Summary	The user enters email and password and after verification will be taken to the homepage.
Pre-Conditions	The user must have an entry in the database and must not be logged in.
Post-Conditions	The user is redirected to the homepage.
Special Requirements	None
Basic Flow	
Actor Action	System Response
1. The user opens the login page	2. The login page is displayed containing a login form
3. The user enters valid email and password	4. Username and password are verified, and the user is taken to the homepage.
Alternative Flow	
5. The user enters invalid email and password	4-A. An error is displayed, and the user is not logged in.

4.6.2 User Logout

Name	User Logout
Actor(s)	User
Summary	A logged-in user logs out of their account.
Pre-Conditions	The user is logged in.
Post-Conditions	The user is redirected to the login page.
Special Requirements	None
Basic Flow	
Actor Action	System Response
1. The user clicks the 'Logout' button	2. User is logged out

4.6.3 User Registration

Name	User Registration
Actor(s)	User
Summary	The user enters a new email and password and after verification, will create an account
Pre-Conditions	The email is not used previously registered
Post-Conditions	A new account is created
Special Requirements	None
Basic Flow	
Actor Action	System Response
1. The user opens the signup page	2. The signup page is displayed containing a login form
3. The user enters a new email and password	4. The email and password are entered into the database and user is taken to the homepage
Alternative Flow	
5. The user already exists	4-A. An error is displayed, and user is not created.

4.6.4 Trigger Alert

Name	Trigger Alert
Actor(s)	User
Summary	If the user sits with wrong posture for longer than 5 minutes, a warning is triggered.
Pre-Conditions	The user must be logged in and the camera must be on.
Post-Conditions	A warning is displayed.
Special Requirements	Camera must be connected.
Basic Flow	
Actor Action	System Response
1. The user sits with wrong posture	2. A 5-minute timer is started
3. Timer ends	4. A warning is triggered telling user to fix their posture
Alternative Flow	
5. The user corrects their posture before timer ends	4-A. Timer resets

4.6.5 Generate Report

Name	Generate Report
Actor(s)	User
Summary	A report is displayed showing different types of statistics
Pre-Conditions	The user must be logged in.
Post-Conditions	Report is displayed
Special Requirements	None
Basic Flow	
Actor Action	System Response
1. The user clicks the generate report button	2. The report is displayed
Alternative Flow	

4.6.6 Edit Timer

Name	Edit Timer
Actor(s)	User
Summary	The user can change the time it takes to trigger the alert.
Pre-Conditions	The user must be logged in and clicks the edit timer button.
Post-Conditions	Timer is updated.
Special Requirements	None
Basic Flow	
Actor Action	System Response
1. The user clicks the settings button	2. The settings page is displayed
3. The user clicks the edit timer button	4. The edit timer page is displayed
5. The user updates the timer within the accepted range	6. Timer is updated
Alternative Flow	
7. The user enters a timer outside the accepted range	6-A. Error displayed

4.6.7 Edit User Information

Name	Edit User Information
Actor(s)	User
Summary	The user can edit personal information such as name, email, password, and other profile details.
Pre-Conditions	The user must be logged in.
Post-Conditions	The user's information is updated in the database.
Special Requirements	None
Basic Flow	
Actor Action	System Response
1. The user navigates to the profile page	2. The profile page is displayed
3. The user selects "Edit Information"	4. Editable fields for name, email, and password are displayed
5. The user modifies their information	6. The system verifies the inputs and updates the information in the database
7. The user clicks "Save"	8. Confirmation message is displayed that the information has been updated successfully
Alternative Flow	
5-A. The user enters invalid information (e.g., existing email)	6-A. An error message is displayed, and the user is prompted to correct the input

4.6.8 Show Homepage

Name	Show Homepage
Actor(s)	User
Summary	The system displays the homepage with an overview of available features after the user logs in or visits the site.
Pre-Conditions	The user must be logged in or accessing the system for the first time.
Post-Conditions	The homepage is displayed with relevant user-specific content or general information for first-time visitors.
Special Requirements	None
Basic Flow	
Actor Action	System Response
1. The user logs in or accesses the website	2. The homepage is displayed with features such as navigation menus, user profile, and system functionalities
3. The user can navigate to other parts of the website	4. Relevant content or features are shown based on the user's role or previous interactions
Alternative Flow	
1-A. The user is not logged in	2-A. The system shows a public login page and an option to log in or sign up

4.7 Hardware and Software Requirements

List of hardware and software requirements for MoveSure.

4.7.1 Hardware Requirements

- Camera: Minimum 480p with 30 FPS
- Server: Core i5 with a basic GPU with 8GB of RAM and 256GB SSD
- Client: Standard laptop

4.7.2 Software Requirements

- Frontend: React, React Native, Angular
- Backend: Python, OpenCV, TensorFlow, Java
- Database: MongoDB

4.8 Graphical User Interface

This section should give the GUI dumps of each screen, with reference to the users. The navigation flow of each user is also required, and each GUI should mark the functionality/use case that it covers.

4.8.1 Sign up Page

This is the page the user sees when they first open the website and it covers Use Case 3

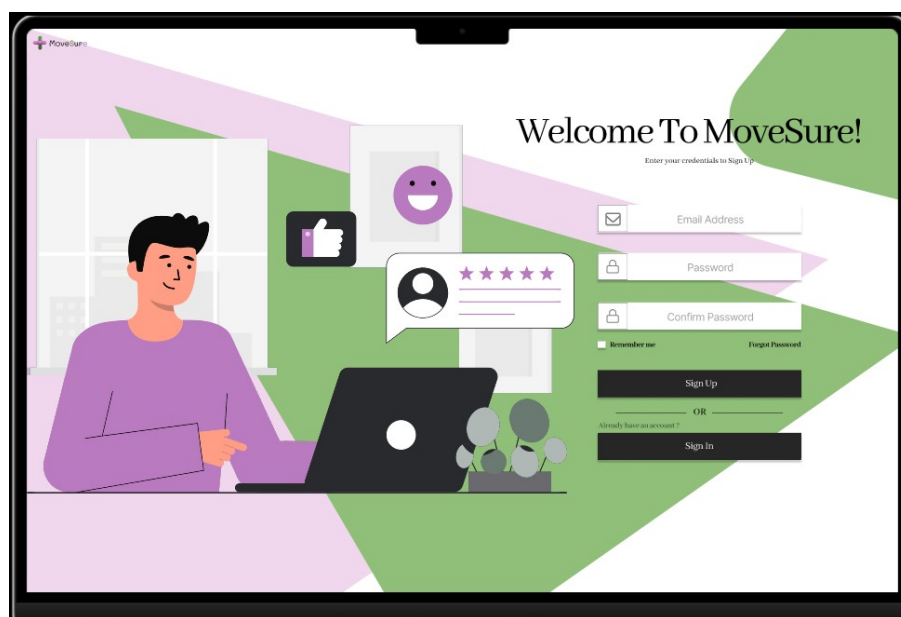


Figure 4.1: Sign up Page

4.8.2 Sign in Page

This page comes up when the user clicks sign in and it covers Use Case 1

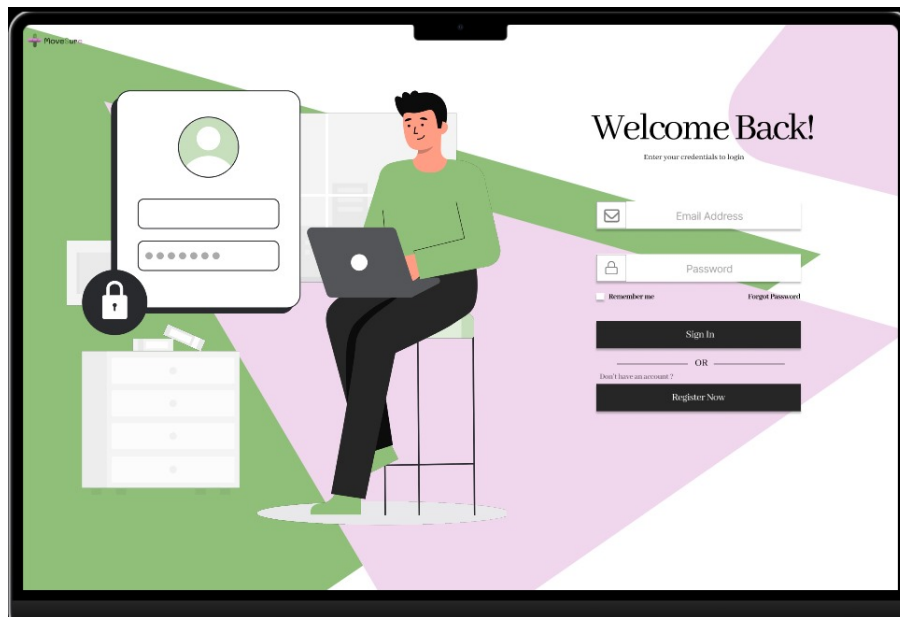


Figure 4.2: Sign in Page

4.8.3 Home Page

Users see the following page after they successfully sign in and it covers Use Case 8

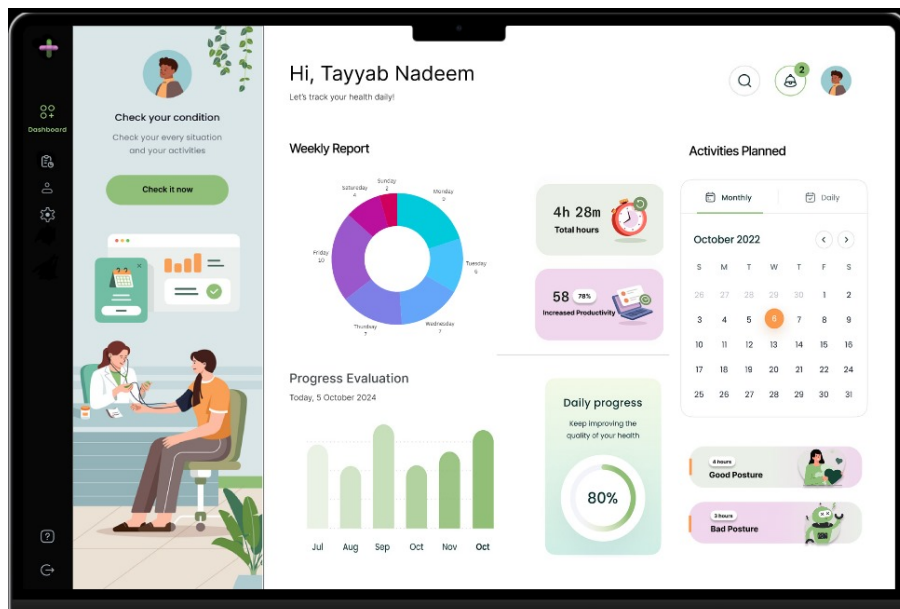


Figure 4.3: Home Page

4.8.4 Report Page

Users can reach this page using the vertical navigation button on the left and it generates reports as shown in Use Case 5

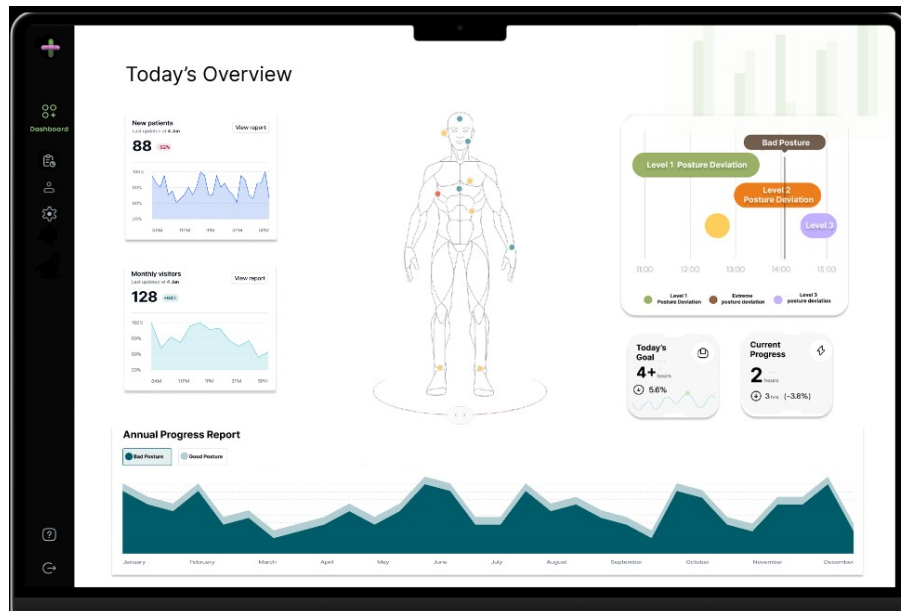


Figure 4.4: Report Page

4.8.5 Alert Notification

Users see this notification once the conditions in Use Case 4



Figure 4.5: Alert Notification

4.8.6 Profile Page

This page is also accessible from the navigation buttons and it also allows the users to edit their information as covered in Use Case

The screenshot displays a user profile page for Tayyab Nadeem, identified as a Software Engineer with Employee ID # 241113. The page is divided into several sections:

- Header:** Displays the user's name, title, and ID.
- Profile Card:** Features a circular profile picture of a person in a green shirt and blue pants, with the name 'Tayyab Nadeem' and title 'Software Engineer' below it.
- Details Section:** Contains fields for Display name (Tayyab Nadeem), Date of Birth (01/08/2001), First name (Tayyab), Last name (Nadeem), and Gender (Male).
- Contacts Section:** Includes fields for Email (tayyabnadeem3037@gmail.com), Mobile number (+92 000 0000000), and Location (somewhere, where nobody cares).
- Password Section:** Provides fields for Enter Current Password, Enter New Password, and Confirm New Password, with an Update Password button.
- Notification Section:** Offers options to manage alerts and updates, with a Learn more link.
- Footer:** Includes a Save Changes button and a disclaimer: 'Remember, all the updated information will be changed in 24 hours after approval by admin office. An email will be sent for authorization.'

Figure 4.6: Profile Page

4.9 Database Design

ER diagram and table of data dictionary.

4.9.1 ER Diagram

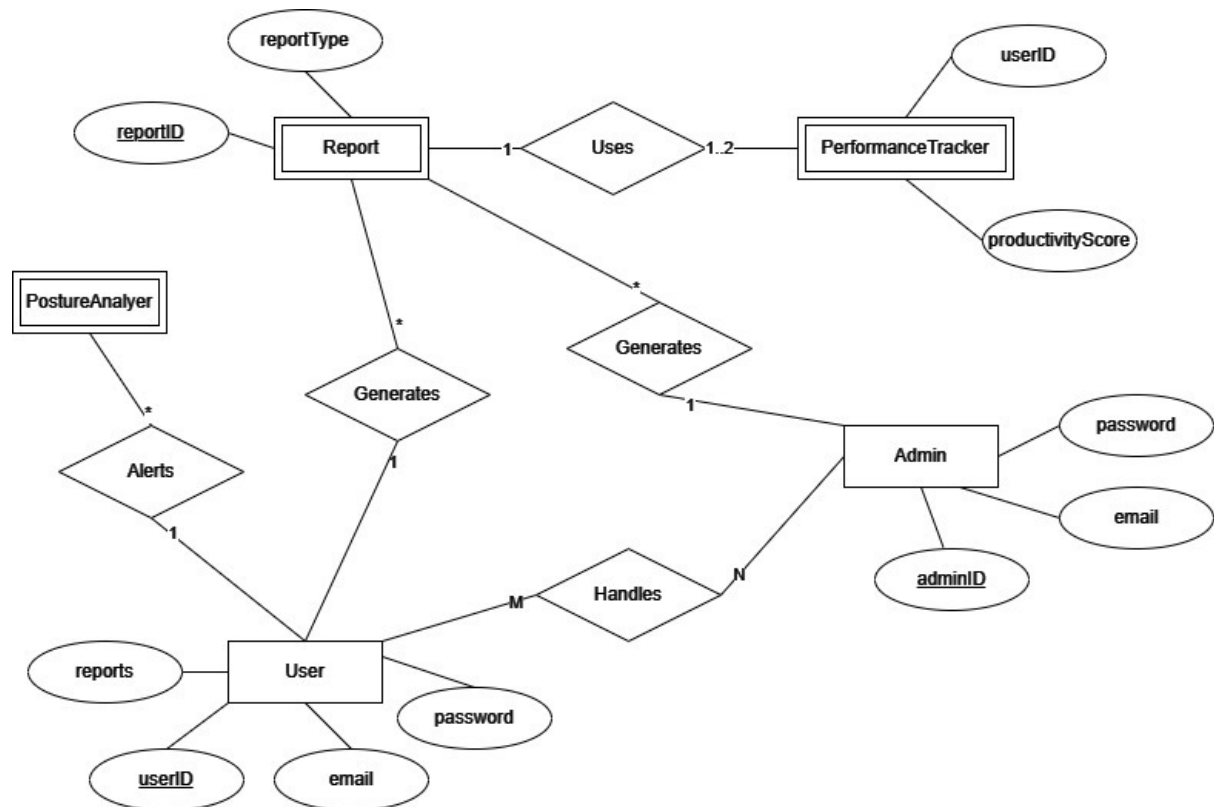


Figure 4.7: ER Diagram

4.9.2 Data Dictionary

Table 4.1: Data Dictionary

Entity	Attribute	Datatype	Relation To	Nullable	Description
User	userID	int	Performance-Tracker	No	Primary Key
	Email	String	-	No	Email of user
	Password	String	-	No	Password of user
	Reports	Report	-	Yes	Previously stored reports of user
Admin	adminID	int	-	No	Primary Key
	Email	String	-	No	Email of admin
	Password	String	-	No	Password of admin
Report	reportID	int	-	No	Primary Key
	reportType	String	-	No	Company report or User report
Performance-Tracker	userID	int	User	No	ID of user whose performance is tracked
	productivityScore	String	-	No	Measure of productivity

4.10 Risk Analysis

A project such as MoveSure presents significant risks: the accuracy and real-time performance of the posture detection system, particularly in a low lighting setup or if the camera has been positioned untimely, and may lead to incorrect posture detection. The other risk is associated with user data privacy during the processing of sensitive data on posture and movement. The system also needs to anticipate high traffic or system load when large organizations are utilizing the system at the same time, which can create potential problems such as performance issues. Another potential challenge includes Internet dependency: breaching of real-time feedback and monitoring due to connectivity interruptions. Furthermore, there is a risk in that users will not interact with the system often or adhere to the proposed recommendations on how to enhance their posture, which might limit the effectiveness of the system. Integration of the system across different environments such as offices, schools, and homes also poses

risks as every place might present various challenges by requiring one to place cameras and enforcing compliance from users in different ways. Finally, the system may still be vulnerable to API failures or hardware malfunctions, and thus in such instances, data will not be processed accurately enough to earn the confidence of the users.

Chapter 5 High-Level and Low-Level Design

This chapter will discuss the high-level and low-level design for our application. In the high-level design, we will focus upon the overall architecture and key modules in relation to other constituents. The low-level design elaborates on the individual modules, classes, and methods. It will incorporate how data structures and algorithms will be implemented in these modules. Both the high-level and low-level design plays a critical role in developing our application MoveSure.

5.1 System Overview

MoveSure is a posture and movement monitoring system, based on artificial intelligence, designed to optimize ergonomics, reduce the risks of musculoskeletal disorders, and improve the general well-being of people in sedentary environments; these include offices, schools, and healthcare settings. Through cameras and AI algorithms, MoveSure provides posture tracking in real time, identifies faulty postures, and detects inactivity with the aim to provide personalized information that could help correct users' posture and support healthy habits throughout the day. The constituent modules are described in the following sections:

5.1.1 Posture Detection Module

This module utilizes AI-powered algorithms to analyze in real time the posture of the users by processing video feeds from cameras attached to it. The module will continuously monitor body movements and pinpoint any deviation to a correct posture. It makes a determination of the posture: correct or incorrect, based on pre-defined ergonomic standards. This module ensures that posture detection is accurate, responsive, and adaptive to individual users.

5.1.2 Inactivity Detection Module

This module tracks the level of activity of the user and detects extended periods spent idle, which lead to bad posture and unhealthy living. Movement tracking with configurable inactivity thresholds generates alerts for when the user has been inactive for quite some time. The module encourages users to take breaks, move around, or adjust their posture to prevent discomfort or strain.

5.1.3 Alert and Feedback Module

MoveSure gives immediate feedback through visual, audio, or text alerts the moment incorrect posture or inactivity is detected. This module allows personalization of the types of and timing of the alerts

depending on user preferences. The module also offers actionable advice about how to improve posture so users can make adjustments on the spot. The module keeps a history of alerts for later analysis in the reporting module.

5.1.4 Reporting and Analytics Module

This module creates extensive reports regarding posture and movement patterns over a of time, and each user has reports made available to them to monitor habits and improvements over time. For teams, administrators will have access to aggregate reports that help in assessing posture health at an organizational level. The reports offer trend analysis and insights into areas where users need to focus on posture correction.

5.1.5 UI Module

The User Interface module offers a very user-friendly dashboard by which people can interact with the system. It can display posture information, alerts, and feedback in real-time and in a clear manner for the user. Users can also set their alert preferences and inactivity thresholds. The UI module supports both individual users and administrators with custom views to monitor and manage the system effectively.

5.1.6 Data Storage and Management Module

The module for data storage shall safely store video data, posture logs and other user preferences. It retains posture reports and allows the readiness of the data for report analysis and historical purposes. The module manages policies over its data retention; it will automatically archive or delete data, according to organizational needs and as it turns out to meet all those compliance requirements in regard to privacy, for example, GDPR regulations.

5.1.7 Security and Privacy Module

The module will take care of all that sensitive input data about video feeds, posture logs, and reports fed into it. Its purpose is to realize a RBAC, which limits access by authorized users who may view the data to guard user privacy. The module further is GDPR compliant and provides features for anonymized data or management of user consent.

5.1.8 Integration and API Module

This module allows third-party application integration, including corporate wellness platforms, health monitoring tools, or any wearable device. It provides RESTful APIs through which other systems can

receive posture and movement data or modify system configurations. The module allows for easy integration with other tools toward expanded usage, such as within HR systems or healthcare platforms.

5.1.9 Hardware Module

This module manages the connected hardware, including cameras, ensuring they function optimally. It provides camera control for calibration and monitoring of system health. The module is designed to support various types of cameras, ensuring compatibility across different hardware configurations.

5.2 Design Considerations

This part explains the findings from consideration of aspects which need to be examined in detail before making the design of the MoveSure system final. These are critical factors so as to guarantee ease of use, performance of the system and changeability in the scenarios in which it will be used.

5.2.1 Assumptions and Dependencies

The design of MoveSure system relies on a few key assumptions and dependencies:

5.2.1.1 Related Software and Hardware

It is taken for granted that the users will be equipped with appropriate software and basic hardware, for instance, personal computers or mobile devices with internet. Further, users will have to be equipped with internet which will be used for data upload and report printing. Cameras meant for posture detection have to comply with certain minimum requirements such as resolution of 1080p, and users' desktop environments should provide up to date web browsers such as Google Chrome, Mozilla Firefox or Safari in order to work with a user interface.

5.2.1.2 Operating System

It provides support for all operating systems like Windows, macOS and Linux as well as mobile operating systems such as Android and iOS. The system is supposed to perform uniformly and reliably in all these operating systems in support of cross-device usage by the end users.

5.2.1.3 End-User Characteristics

For vendors, the target users are likely to possess a certain level of technology acceptance such as web surfing, basic settings of the system etc. Furthermore, organizations that will carry out MoveSure will

also have administrative personnel that can perform user management, set up configurations of system settings, and read analytical reports. Advanced proper use of any device is expected of the users.

5.2.1.4 Possible and/or Probable Changes in Functionality

As with other fields, ergonomics does not remain static and new best practices are bound to be developed. The system has been made in such a way that it can accept changes in posture detection systems, recommendations of ergonomic experts, and even connectivity with new health monitoring devices in the future. Integration of other advancements including the use of wearable devices or advancement of software embedded in the device can also be thankfully included in future advancements.

5.2.2 General Constraints

List of constraints that this project will experience.

5.2.2.1 Hardware or Software Environment

The achievement of this level of posture monitoring depends largely on the hardware available for use, especially the cameras and the processing units. The system is based on the availability of good quality cameras (minimum 1080p resolution), a good internet connection, and good processing units (any intel Core i5 or NVIDIA GTX 1650 and above) to facilitate the feedback and reporting in real time. The performance of the hardware in use at any of the locations will affect the usefulness of the system.

5.2.2.2 End-user Environment

The set up will range from corporate offices and schools to medical centers. It is designed to ideal workplace conditions with respect to the internal lighting and the position of the camera aimed at making it possible to detect postures accurately. MoveSure's efficiency relies on how well its users understand web applications, and it is expected that all users will be working in a controlled environment, favorable for proper posture monitoring.

5.2.2.3 Interoperability Requirements

It is foretold that MoveSure will cooperate smoothly with present hardware such as cameras, sensors and external symbiotic systems such as health monitoring and corporate wellness applications. The architecture of the system contains capabilities for creation of applicative programming interfaces (API) with external systems that enhance the insightful reporting within the system.

5.2.2.4 Protocol Requirements

The system must follow the requirements of standard web applications where all the data transferred across the systems by the internet are encrypted through HTTPS. It utilizes HTML5, CSS3, and JavaScript as the front end programming languages to create an interactive interface. The interface is expected to be user-friendly and make it possible to operate the system on desktop computers, tablets, and smart phones for uniform interaction.

5.2.2.5 Requirements concerning Data Repository and Distribution

The system features data storage containing posture logs, video streams, and analytical reports. Given the nature of this information, it has to be organized within a reliable, and highly-scalable database that can efficiently accommodate it. The system incorporates either cloud or on-premise storage solutions based on the needs of the organization. The data repository must also allow such operation as data retrieval and access for the timely generation of reports.

5.2.2.6 Requirements concerning Verification and Validation

Due to the nature of the system, which involves dealing with sensitive information especially for health objectives, it is mandatory that verification and validation is in place. The algorithms of posture detection should be put through more than reasonable trials, the validation of the system on error responses especially in real time operations, has to be done. The data security features also need to be validated, especially regarding data privacy laws such as GDPR.

5.2.3 Goals and Guidelines

The major characteristics that in a way control the design and the substance of MoveSure are as follows:

- **KISS (Keep It Simple, Stupid):** The system design needs to be kept low in complexity so as to avoid frustration among users as they try to locate the interface elements, change the configurations or run the available reports.
- **User-Centric Design:** Such systems should aim at enhancing the user experience through ease of interaction, readability, and accessibility of the system interface. It will horizontally be developed to target users and non-users of advanced ergonomics and posture monitoring tools across different profiles.
- **Flexibility and Adaptability:** MoveSure device has to be flexible enough to accommodate such changes as the addition of other more updated applications within the system including Advanced

AI algorithms, new hardware sensors and new ergonomic accessories, and smartphones. It has to transcend the modular design towards a deep understanding of the user and the ergonomics.

5.2.4 Development Methods

It is essential to note that MoveSure will be using an agile methodology along the entire development lifecycle so as to be able to adapt and respond to feedback during the development process. The development process will incorporate the following methods:

- **Agile Methodology:** At every stage and particularly during system/sprint reviews, the system improvements will be made taking into consideration any changes in user needs and their interaction with the product, and improvements in ergonomic practices in general and that of posture systems in particular that happen in between system development sprints.
- **Scrum Framework:** Usage of the Scrum framework will facilitate the organization of all development processes and hence emphasizes on people working together in small teams, having small, frequent meetings to track progress and improve the result step by step. Sprints will provide a stage for the features like the accuracy of the posture detection feature, the decision-alert notification system customization, and reporting tool to be built.
- **Continuous Integration and Continuous Deployment:** The system will employ CI/CD methodologies due to the fact that new features, updates, and patches to the system will be adopted without leading to a change in end-users, thereby enhancing the user experience. Such methods will help in ensuring that MoveSure is flexible and easy to use, as well as technologically competent to address users needs as well as the enhanced features of posture monitoring and ergonomics in the future.

5.3 System Architecture

The MoveSure system is a posture monitoring and feedback system that integrates various recent posture monitoring technologies and frameworks. The architecture encompasses internal module design and external system interfaces so that all modules are integrated. The structure of the system is also presented in diagrammatic form.

5.3.1 High Level Partitioning

The system comprises five (5) main components. These components can be deployed either in a networked computer system or in a stand-alone computer depending on the choice.

- **Presentation Layer:** In the layer systems the interface is given and developed by modern development web technologies such as React.
- **Business Logic Layer:** A separate web interface built on a strong framework like Django or Flask takes care of the application logic, user sessions, and communication of various modules.
- **Data Persistence Layer:** Information regarding posture logs, user profiles, and alert histories is stored using a RDBMS such as MongoDB or MySQL in order to retrieve such data in a non-permanent memory.
- **Integration Layer:** This is the component which connects to the AI algorithms and posture detection models and handles the real time analysis of posture and movement data with Tensorflow or Pytorch.
- **Hardware Management Layer:** Such interactions with the physical devices such as cameras and sensors are managed by this module to ensure that these devices are interfaced properly with the system.

5.3.2 Component Collaboration

The high-level components of the MoveSure system collaborate in a structured manner to ensure seamless functionality. The Presentation Layer interacts with the Business Logic Layer through the use of RESTful APIs in which data is sent and received when users send commands or alerts. The application logic and incoming data processing as well as communication with the Data Persistence Layer in regards to data storage requires work from the Business Logic Layer. The annotation layer uses the AI models for posture recognition by processing images from the Business Logic Layer and provides processed information for user interaction. Also, the Hardware Management Layer is also responsible for establishing the connection with the cameras and sensors and transmits the information captured by them to the Business Logic Layer for further processing. The database is an organizational unit that maintains all the system's information with the respect of reliability and security, however, it also provides analytical reporting for trends or behaviors on the usage of the application.

5.3.3 Rationale for Decomposition

This type of architectural design offers a degree of scalability, maintainability, and effective encapsulation of different functional requirements. Each component is self-contained in the sense that it can be developed, tested and maintained individually eliminating delays in the introduction of new features or updates. The core structure in addition supports the requirements of the system, that is, the need for other components and processes to progress in real time with the end user experience. Other options,

like building it as one block a monolithic architecture, were evaluated, however, the existing structure was selected because of the ease of adding AI features and hardware connections.

5.3.4 System Architecture Diagram

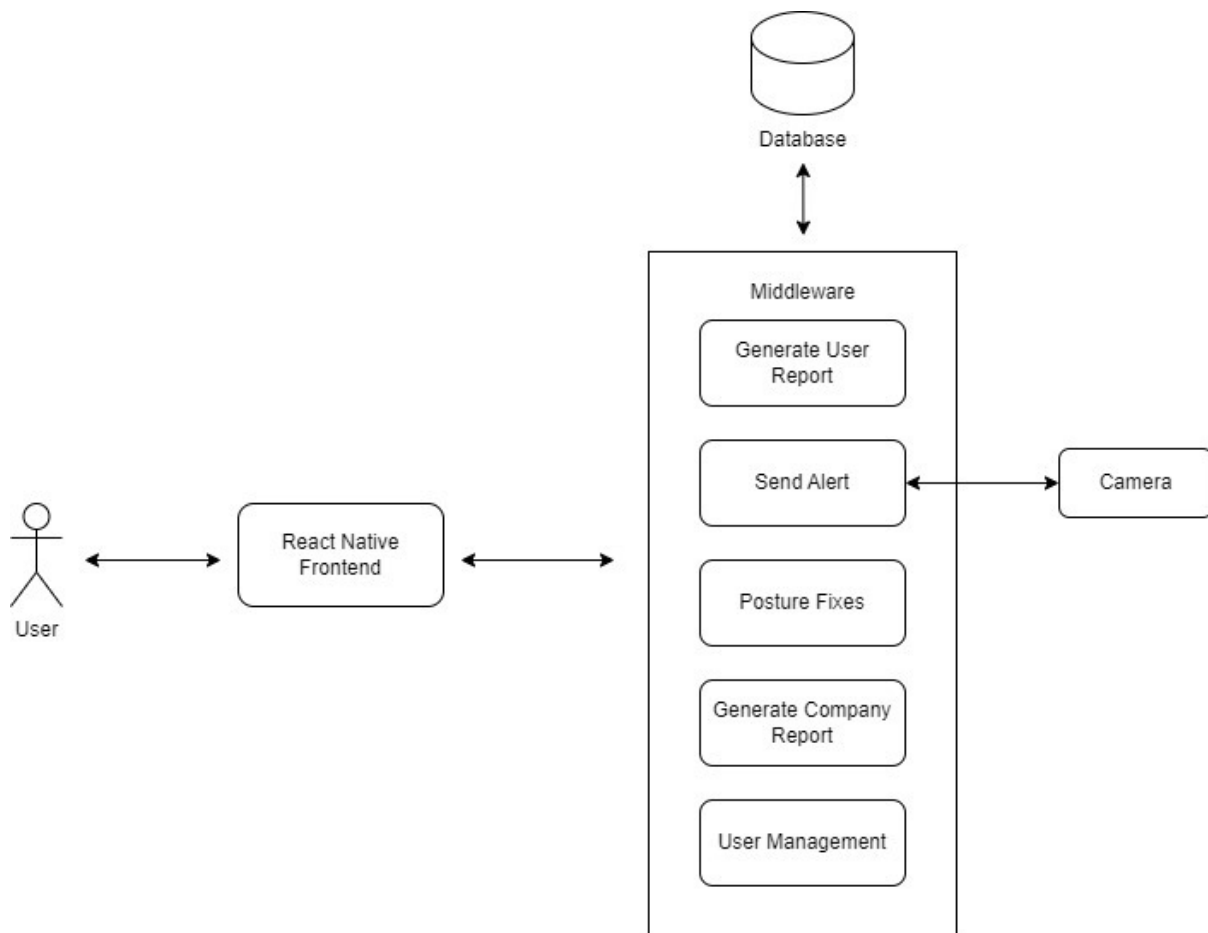


Figure 5.1: Architecture Diagram

5.3.4.1 High Level System Architecture Diagram

The major layer out of the high level architecture diagram to the lowest level right has been shown out. Depending on the focus on primary functions of the major layers the constituent elements-wise Logical Interconnection diagrams descriptions, features, data flow and controls, and sinkhole system communication with other systems are depicted. The process of data processing and user interactions and feedback of the system are shown and how the responsiveness and effectiveness of the system is sustained.

5.3.5 Subsystem Architecture

List of layers in the subsystem architecture.

5.3.5.1 Presentation Layer Subsystems

The Presentation Layer consists of further its radius strut:

- **User Interface Subsystem:** This subsystem is responsible for creating an efficient and easy to use interface through which the users can interact with the system. It encapsulates user oriented dashboards which contains real-time posture information, alerts, and reports to the selected users as per their needs.
- **Authentication and Authorization Subsystem:** This subsystem oversees user security and permissions within the application where role based access control is exercised. In this case, the access and the actions that each user is allowed to perform within the system is defined by the user's role.
- **Notification Management Subsystem:** This subsystem is responsible for any alerts and notifications that the users receive in relation to the correction of the users' postures and the users' activity levels, as well as their performance feedback on the vices. It promotes both timeous communication and compliance amongst users.

5.3.5.2 Business Logic Layer Subsystems

The Business Logic Layer consists of the following subsystems:

- **User Management Subsystem:** Subsystem deals with creation of user accounts, setting user preferences and updating profile information about users. It performs user registration, logging in procedures and changing user preferences.
- **Posture Analysis Module:** This module receives information from the systems dealing with detecting postures and studies the posture histogram generated. Based on this understanding, the system provides measures in real time to correct any deficiencies Aimed at Posture Improvement.
- **Inactivity Detection Module:** This module monitors how active the user is and logs periods of inactivity, after which warnings and reminders for the user to move or change their posture are issued when appropriate.
- **Reporting Module:** This module of the system gathers information on the posture and activity of users and administrative staff and generates reports based on this information. It tracks time trends, spotting places needing development and producing a timescale for the data gathered.
- **Integration with AI Models:** This module communicates with AI models regarding the detection of posture and movement analysis. It processes the requests from the models and retrieves results

from them to oversee how the task is being performed.

5.3.5.3 Database Subsystems

The Database Layer design includes mechanisms for data to be stored and managed as follows:

- **Subsystem for Data Storage:** This subsystem is designed to permanently store all user information, posture readings, activity data, and even reports with analyses. This ensures that data is kept safe, and easy to retrieve when need be.
- **Subsystem for Data Retrieval and Query Processing:** This subsystem takes care of the processing of queries regarding posture data and user analytics. This is done with optimum efficiency as specific data is retrieved from the database per a request made by the business logic layer.

5.3.5.4 Hardware Management Subsystems

The focus of this layer is on proper ownership, use and maintenance of devices.

- **Camera Management Subsystem:** This subsystem is effective in monitoring the attached cameras in terms of their setting up and usage. In this context, cameras must be optimally set and calibrated so as their internal systems correctly capture one's posture.

5.3.5.5 Security and Privacy Subsystems

- **Data Encryption Subsystem:** During this process, this subsystem makes certain that every critical information of the users like videos and postures are encrypted at all times over the internet or while being stored to avert hacking.
- **Access Control Subsystem:** This subsystem specifies the privileges and amount of data the user can access within the system working on the privacy as well as security concerning the user's data as appropriate information is restricted to authorized users only.

5.4 Architectural Strategies

This part of the paper explains the factors influencing the structure of MoveSure system and its top levels.

5.4.1 Choice of Technologies

- **React JS for Front-End:** Front End development, the use of React JS has been adopted in the development of the user interface because of excellent responsiveness in the development of web

applications. This perfectly fits within the aim of the MoveSure system which is to enhance the usability of the interface in viewing the real-time posture detection and communication of posture feedback to users.

- **Backend markt Python and Django:** Back-end web development is supported by the use of Python, in particular the use of the Django Web framework as Python is a flexible programming language and Django is a powerful framework. This brings fast system development, integration of AI models for detecting human postures and management of user information as well as reports.
- **MongoDB Database:** Out of the various types of database management systems, the NoSQL type, particularly MongoDB has been adopted due to the reasons of its flexibility, scalability, and effective structuring of unstructured and semi-structured information. It is excellent in the arms of all those kinds of data which in return helps to deal with posture history, activity logs along with insightful reports. In MongoDB, the data is stored in the document model and that facilitates smooth upscaling and quick fetching of the bulk assets that are needed in addressing matters regarding posture.

5.4.2 Reuse of existing Software Components

- **Tailwind CSS:** For the stylistic purposes of this project, in the case of user interface cross-cutting concerns, Desiring for an application that would be made fancy/tailored will require Tailwind CSS to be employed especially in the first stages of incorporating different design compositions.
- **CoreUI react:** The interaction elements of the application shall be designed based on the core ui design system and its components and templates to ensure design consistency and user satisfaction within the application's interfaces.

5.4.3 Future Plans for Extension or Enhancement

The MoveSure is built with a highly modular and highly scalable architecture which accommodates additional developments and changes that take place in the subject of ergonomics and health tracking devices. The software design principles, particularly the SOLID principles, will guide the development process, with a focus on the Open/Closed Principle to allow for extension without modification of existing code.

5.4.4 Version Control

The MoveSure application shall be handled as a set of gists in one github. The code will be changed through the pull request system and no less than two teammates will have to approve such changes. This

procedure enhances the development lifecycle as well as guarantees that every piece of code is always reviewed, and proper coding standards are followed.

5.4.5 User Interface Paradigms

The system will place great emphasis on providing a user interface that will be easy to learn and responsive through the use of React JS and Tailwind CSS. The principles of interface design will comply with Shneiderman's Eight Golden Rules of Interface Design.

5.4.6 External Database and Data Storage Management

PostgreSQL technologies will fully accommodate the corpus of storing and managing data pertaining to the MoveSure system in such a way that data is preserved, secured, and scaled up in case polymorphic constellations of posture and activity data increases with time.

5.4.7 Concurrency and Synchronization

To enhance user experience, asynchronous operations will be utilized on the front end, allowing users to interact with the website while the system processes time-consuming tasks, such as web API requests. On the backend, parallel processing will be implemented to ensure the application can handle multiple users simultaneously. Django and Python's built-in mechanisms will efficiently manage concurrent user interactions and system processes, maintaining performance under load.

5.5 Domain Model/Class Diagram

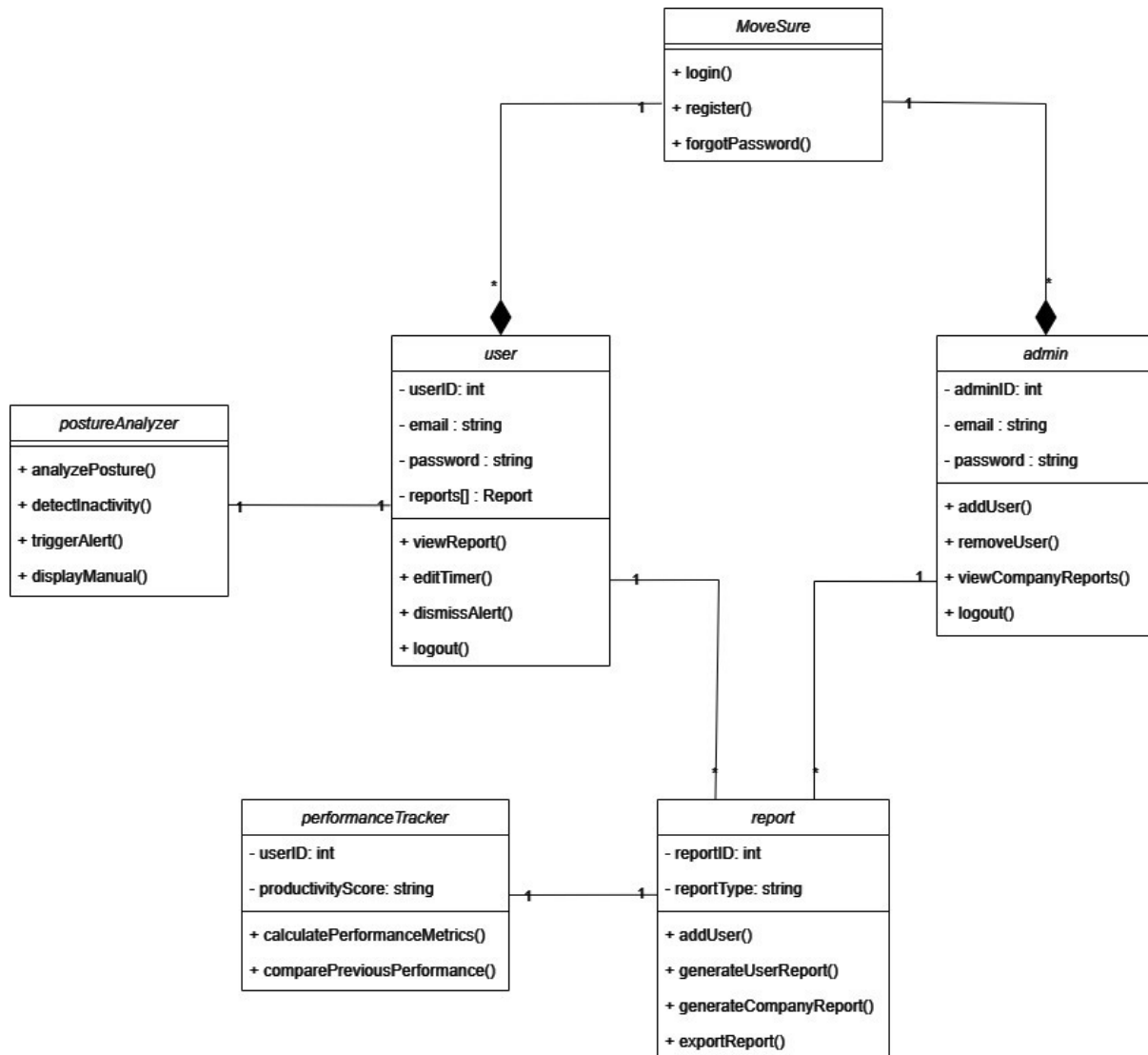


Figure 5.2: Class Diagram

5.6 Policies and Tactics

In order to maintain the quality and effectiveness of MoveSure application, the project adopts different practices and procedures that are common in the industry. The following policies are some of the guiding policies that will be observed in the course of developing the system.

5.6.1 Coding Guidelines and Conventions

The MoveSure project will follow particular coding practices to enhance adherence to appropriate structure such as formatting, congestion, and other important emulsion characteristics. Back-end develop-

ment will conform to PEP 8 guidelines while JavaScript ES6 standards will be used in front-end development. Best practices adhered to in the industry will also be employed with regards to documentation, code quality and cleanliness. Choice of IDEs for projects undertaken in the classroom remains to be Visual studio code, as it is light in terms of runtime and cross platform.

5.6.2 Testing Strategies

After the review, acceptance, unit integration and system testing has also been accomplished, functional and non-functional testing of the application will be carried out. A white rectangular testing measure and black rectangular testing will be engaged. For black box testing, decision tables will be utilized while UI testing, API testing and unit testing will be part of the white box testing regimes to ensure that the components within the application function as intended.

5.6.3 Continuous Integration and Deployment

The frequency of code checking and automated testing or deployment will be aided by the creation of Continuous Integration/Continuous Deployment (CI/CD) pipelines. The code is expected to be hosted on GitHub, and the secondary development tools to be developed, will include frequent syncing or merging of new code changes into the main branch after valid approval and review of the developed new codes.

5.6.4 Source Code Organization

The source code for MoveSure shall be structured in modules and directories according to the functions hierarchy. This will improve organization of code and relieve the burden of orientation to the members of the team who will shrink into the codebase.

5.6.5 Error Handling and Recovery

The error handling and log capturing mechanisms will be vigorous so that the system can remain intact and bounce back to normal even after a system failure. This would include the collection of the captured error logs as well as the implementation of error messages which will be extended to the end users.

5.6.6 User Interface Design

Reactions and the use of Tailwind CSS will employ the principles of responsive design to ensure that the end users adapt to different environments without interruptions. The web application will be made adaptive towards all suns as well as application screens to ensure that users' experiences are as uniform as possible no matter the smartphone used to log in into the application.

5.6.7 Database Management

With the use of Django ORM, the application will handle the database without the normal database interactions thereby making the application less prone to SQL injection since there will be less database queries. This will improve application security and manageability of data within the application.

5.7 Conclusion

The MoveSure application is able to provide the degree of control needed for a person using the application and help the user feedback. Oriented only towards user interface and ease of comprehension there is well known modifiable and expandable architecture. Having mentioned so many industry standards and best practices, the effectiveness of the solution developed for the problem of posture restoration and health improvement will be natural.

Chapter 6 Implementation and Test Cases

This chapter discusses the work that have been done by the team so far along with some technical details and back-end modules that were not discussed in the high and low-level design. The content in this chapter is not final and can evolve as we go on with implementing more modules.

6.1 Implementation

During the project's implementation phase, our team has put a dedicated effort in developing and exploring different tools and utilities that will help us in the further implementation of our application. The team investigated several technologies to construct and improve the functionality of MoveSure. The implementation details of the work done so far are below:

6.1.1 User Registration

The Registration Module expects an input of basic user details, like name and preferences. It associates that input of basic information with posture tracking settings for that particular user. Therefore, facial recognition is not what the module necessitates but ties all unique profiles of individuals with a particular posture monitoring in the said module. Data files are saved safe in one of the two set-up databases: Firebase or MySQL.

6.1.2 User Authentication

The Authentication Module authorizes secure user access by authenticating their credentials-for example, login, username, and passwords. It loads personalized settings regarding posture when successfully authenticated at login, thus enabling a bespoke experience. Firebase Authentication will take care of the login since it ensures secure authentication in a reliable manner for all users.

6.1.3 Profile Customization

Personal experience is made possible by having customizability through the profile Customization Module. Users personalize setting preferences, such as being reminded about posture, setting their preferred feedback type from various sources, and updating their inactivity tracking to threshold values. All are saved to a Firebase or MySQL backend database, making personalization easier for users according to their preferences. The backend has been designed to be user-friendly, resulting in the use of React or Flutter for comfortability.

6.1.4 Start/Stop Monitoring

This module provides a simple on/off button that can easily activate or deactivate posture monitoring. It clicks the system to turn on real-time monitoring of the user's posture using a webcam and display real-time feedback regarding their posture and activity. The function of buttons is incorporated into the HTML5 and JavaScript frontend to enable users to gain quick interaction and control.

6.1.5 Generate Reports

The Feedback Report Module provides detailed reports based on user activity in terms of session length, posture habits, and instances of inactivity. These reports allow the users to monitor their improvement with regard to posture. To make this data exciting, Chart.js or D3.js can be used for interactive and visually attractive graphical reports that can help the user understand his posture trends over time better.

6.1.6 Fetch Data from the Database and Camera Module

This module tracks the user's posture data by continuous scanning in the background and updates user data stored in the database with real-time camera inputs from the user. All of the user's activities along with their progress data is updated, retrieved, and stored using a Firebase or MySQL database to ensure data consistency with real-time updates. To process the camera feed in order to track posture, OpenCV is utilized.

6.1.7 Identifying Individual

The system monitors posture data and correlates it with a unique user profile based on the data recorded during the registration process in this module. A unique identifier will be used to correlate the user's data with his/her profile so that all feedbacks and reports are accurate and personal. Firebase or MySQL databases are used to store and link this data securely.

6.1.8 Generating Alerts

The Alert System is a vital feature that notifies the user to monitor their posture and movement throughout the day. It reminds the user gently when it notices that their posture is incorrect or that they have been sitting in the same position for an extended period of time. The real-time feedback assists the user in sitting straight, encourages healthier behaviors, and avoids discomfort. This enables users to make prompt corrective measures to enhance their ergonomic patterns. The alerts are integrated directly with the real-time feeds generated from the camera, analyzed using OpenCV, and securely stored in the backend databases, e.g., Firebase or MySQL, to provide an uninterrupted and responsive experience.

6.2 Test Cases

This section outlines detailed test cases designed to validate each core use case of the MoveSure system.

Table 6.1: User Login

User Login			
UC-01			
Test Case ID:	TC-UL-001	QA Test Engineer:	Muhammad Abdullah
Test Case Version:	1	Reviewed By:	Aaon Raza
Test Date:	March 30, 2025	Use Case Reference(s):	UC-01: User Login
Revision History:	None		
Objective:	To verify that the system logs in a user with valid credentials and redirects to the Homepage.		
Product/Ver/Module:	MoveSure Web Application		
Environment:	Standard desktop environment with a modern browser.		
Assumptions:	User is pre-registered and network connectivity is stable.		
Pre-Requisite:	Login page is accessible.		
Step No.	Execution Description	Procedure Result	
1	Navigate to the Login page.	Login page is displayed.	
2	Enter valid username and password, then click the Login button.	System validates credentials and redirects to the Homepage.	
Comments: Error message is displayed for invalid credentials.			
Passed			

Table 6.2: User Logout

User Logout			
UC-02			
Test Case ID:	TC-ULOG-001	QA Test Engineer:	Aaon Raza
Test Case Version:	1	Reviewed By:	Tayyab Nadeem
Test Date:	March 31, 2025	Use Case Reference(s):	UC-02: User Logout
Revision History:	None		
Objective:	To verify that a logged-in user can successfully log out and be redirected to the Login page.		
Product/Ver/Module:	MoveSure Web Application		
Environment:	Standard desktop environment with a modern browser.		
Assumptions:	User is already logged in and session management is working properly.		
Pre-Requisite:	Homepage with a visible Logout button is displayed.		
Step No.	Execution Description	Procedure Result	
1	Click the Logout button on the Homepage.	System ends session and redirects to Login page.	
Comments: Auto-logout due to inactivity should prompt a warning before logout.			
Passed			

Table 6.3: User Registration

User Registration			
UC-03			
Test Case ID:	TC-UR-001	QA Test Engineer:	Tayyab Nadeem
Test Case Version:	1	Reviewed By:	Muhammad Abdullah
Test Date:	April 1, 2025	Use Case Reference(s):	UC-03: User Registration
Revision History:	None		
Objective:	To ensure that a new user can register successfully with valid details and receive a confirmation.		
Product/Ver/Module:	MoveSure Web Application		
Environment:	Standard desktop environment with a modern browser.		
Assumptions:	All mandatory fields are completed with valid data.		
Pre-Requisite:	Registration page is accessible.		
Step No.	Execution Description	Procedure Result	
1	Navigate to the Registration page.	Registration page is displayed.	
2	Enter all required details (username, email, password, etc.) and click Submit.	System validates and stores the new user details, showing a confirmation message.	
Comments: Validation errors should appear for invalid entries (e.g., invalid email or mismatched passwords).			
Passed			

Table 6.4: Trigger Alert

Trigger Alert			
UC-04			
Test Case ID:	TC-TA-001	QA Test Engineer:	Muhammad Abdul-lah
Test Case Version:	1	Reviewed By:	Tayyab Nadeem
Test Date:	April 2, 2025	Use Case Reference(s):	UC-04: Trigger Alert
Revision History:	None		
Objective:	To ensure the system detects an incorrect posture and immediately triggers an alert.		
Product/Ver/Module:	MoveSure Posture Monitoring Module		
Environment:	Standard desktop with webcam in an adequately lit room.		
Assumptions:	User is logged in and monitoring is active.		
Pre-Requisite:	Posture detection module is operational.		
Step No.	Execution Description	Procedure Result	
1	Simulate a poor posture by adopting an incorrect sitting/-standing position.	System captures the posture and identifies deviation.	
2	Wait for the system to trigger an alert.	Alert (visual/audio) is triggered immediately.	
Comments: System should avoid false alerts.			
Passed			

Table 6.5: Generate Report

Generate Report			
UC-05			
Test Case ID:	TC-GR-001	QA Test Engineer:	Aaon Raza
Test Case Version:	1	Reviewed By:	Muhammad Abdullah
Test Date:	April 3, 2025	Use Case Reference(s):	UC-05: Generate Report
Revision History:	None		
Objective:	To confirm that the system generates and displays a detailed posture report from accumulated user data.		
Product/Ver/Module:	MoveSure Reporting Module		
Environment:	Standard desktop environment with a modern browser.		
Assumptions:	User has accumulated posture monitoring data.		
Pre-Requisite:	Report page is accessible and user is logged in.		
Step No.	Execution Description	Procedure Result	
1	Navigate to the Report page.	Report page is displayed.	
2	Click on the “Generate Report” button.	System retrieves data and displays a detailed posture report.	
Comments: The report should include trend insights and refresh with the latest posture data.			
Passed			

Table 6.6: Edit Timer

Edit Timer			
UC-06			
Test Case ID:	TC-ET-001	QA Test Engineer:	Tayyab Nadeem
Test Case Version:	1	Reviewed By:	Aaon Raza
Test Date:	April 4, 2025	Use Case Reference(s):	UC-06: Edit Timer
Revision History:	None		
Objective:	To verify that the user can modify the alert timer setting and that the change is saved successfully.		
Product/Ver/Module:	MoveSure Settings Module		
Environment:	Standard desktop environment with a modern browser.		
Assumptions:	Timer values must be within defined acceptable ranges.		
Pre-Requisite:	User is logged in and on the Timer Settings page.		
Step No.	Execution Description	Procedure Result	
1	Navigate to the Timer Settings page.	Timer Settings page is displayed.	
2	Enter a new valid timer value and click “Save Settings.”	System validates input, updates the setting, and confirms the change.	
Comments: Test with invalid values should trigger error messages.			
Passed			

Table 6.7: Edit User Information

Edit User Information			
UC-07			
Test Case ID:	TC-EUI-001	QA Test Engineer:	Muhammad Abdul-lah
Test Case Version:	1	Reviewed By:	Aaon Raza
Test Date:	April 5, 2025	Use Case Reference(s):	UC-07: Edit User Information
Revision History:	None		
Objective:	To ensure that the system updates user profile information correctly.		
Product/Ver/Module:	MoveSure User Profile Module		
Environment:	Standard desktop environment with a modern browser.		
Assumptions:	User input meets validation criteria (e.g., proper email format).		
Pre-Requisite:	User is logged in and on the Profile page.		
Step No.	Execution Description	Procedure Result	
1	Navigate to the Profile page and click “Edit User Information.”	Edit form is displayed.	
2	Modify desired fields (e.g., name, email) and click “Save Changes.”	System updates information and shows confirmation message.	
Comments: Invalid input should prevent saving and display errors.			
Passed			

Table 6.8: Show Homepage

Show Homepage			
UC-08			
Test Case ID:	TC-SH-001	QA Test Engineer:	Aaon Raza
Test Case Version:	1	Reviewed By:	Tayyab Nadeem
Test Date:	April 6, 2025	Use Case Reference(s):	UC-08: Show Home-page
Revision History:	None		
Objective:	To verify that the Homepage loads correctly post-login displaying all key sections.		
Product/Ver/Module:	MoveSure Web Application – Homepage Module		
Environment:	Standard desktop environment with a modern browser.		
Assumptions:	Dynamic content (dashboard, alerts, etc.) is available and updated.		
Pre-Requisite:	User is logged in and redirected to the Homepage.		
Step No.	Execution Description	Procedure Result	
1	Verify redirection to the Homepage after login.	Homepage is displayed with all key sections.	
2	Check that dashboard, alert, and dynamic content modules are visible.	All modules display current, user-specific data.	
Comments: All sections must load with up-to-date information.			
Passed			

Chapter 7 User Manual

7.1 Beginning

- Verify that your device has an internet connection and a working camera (at least 480p, 30 frames per second).
- Go to the MoveSure platform by opening your browser.
- If you are not logged in, the login page will be displayed.
- To create an account if you don't already have one, go to the signup page.

7.2 Registering as a User

- On the login page, select the Register Now” link.
- Provide a working email address and password when completing the form.
- To create your account, click Register.”
- An error message will appear if the email address is already registered.

7.3 User Login

- On the login page, type your registered email address and password.
- Press the “Sign In” button.
- You will be taken to the homepage if the credentials are legitimate.
- An error message will appear if it is invalid.

7.4 User Logout

- From the top menu or sidebar, select the “Logout” button.
- You will be taken to the login page at the conclusion of your session.

7.5 Overview of the Dashboard

- The dashboard, or homepage, appears after logging in.
- See your calendar view, daily progress, weekly report, and welcome message.
- Profile settings, analytics, and other pages are accessible through the sidebar.

7.6 Monitoring Posture

- Verify that your camera is turned on.
- Take a regular seat at your desk.
- A warning appears on the screen if poor posture is observed for longer than five minutes.
- Adjust posture before the alert timer resets.

7.7 Create Weekly Report

- Go to the dashboard's Weekly Report area.
- To view a breakdown of good and bad posture, click "Generate Report."
- Productivity data and a pie chart will be displayed.

7.8 Timetable and Development

- From the dashboard's right panel, navigate to the Calendar section.
- Track your progress in posture each day on particular dates.
- Days with colour coding—green for good, red for bad—reflect posture performance.

7.9 Profile Management

- Select the profile icon located in the upper right corner.
- See your name, email address, and usage statistics as of right now.
- To edit your information and save your changes, click "Edit."

7.10 Timer Setting

- Select the “Settings” icon located in the sidebar.
- To modify the alert duration (e.g., 5, 10, 15 minutes), select “Edit Timer.”
- Click “Save” after entering your preferred time.
- An error message appears if the input is invalid (for example, a negative number).

Chapter 8 Conclusion and Future Work

8.1 Conclusion

The MoveSure team was able to deliver the initial implementation of the proposed requirements, meeting all the core objectives of the project. It has been designed after researching and thoughtful design to create a system which uses real-time posture tracking that identifies incorrect posture and delivers timely corrective alerts. Personalized reports will integrate tracking of one's progress, moving users toward healthier ergonomic habits over time. It is developed in the most user-friendly form and intuitive interface, so it is appropriate for a variety of applications, including offices, educational, and home use.

The architecture designed is scalable so that any future enhancement can be added seamlessly. Therefore, its goal is to change users' posture, productivity, and well-being in the long run for healthier working or studying habits.

8.2 Future Work

The team plans to launch new advanced features that include full posture detection, real-time posture and inactivity alerts, and long-term progress. We will also include very detailed reports and step-by-step guidance on how people can correct their posture effectively, which will make MoveSure even more able to manage posture in a more precise, holistic manner to enhance healthy and productive behavior in users.

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